A NON-B-SEMI-AMPLE MODULI PART OF AN LC-TRIVIAL FIBRATION WITH NON-EFFECTIVE BOUNDARY

JIHAO LIU

Abstract. We show that there exists an lc-trivial fibration of relative dimension 1 whose boundary is not effective over the generic point of the base and whose moduli part is not b-semi-ample, so the effectivity hypothesis in the recent b-semi-ampleness theorem for lc-trivial fibrations cannot be removed. We also show that, in the formulation of Han, Jiang, and Luo, which allows non-effective boundaries, Shokurov's conjecture on singularities on the base of a Fano type fibration fails in relative dimension 2.

Contents

1. Introduction 1
2. Preliminaries 5
3. The moduli part as a Hodge bundle 7
4. Period map and monodromy of cyclic covers of elliptic curves 12
5. A nef Hodge line bundle that is not semi-ample 17
6. Proof of Theorem 1.3 20
7. Realizing the Hodge line bundle by sub-pairs on the weighted projective plane 21
8. Proof of Corollary 1.4 26
References 30

1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Let $\pi : (X, B) \to Z$ be an lc-trivial fibration (see Definition 2.4), with discriminant $\boldsymbol{b}$-divisor $\mathbf{B}$ and moduli $\boldsymbol{b}$-divisor $\mathbf{M}$ (see Definition 2.5). Then

$$K_X + B \sim_{\mathbb{R}} \pi^*(K_Z + B_Z + M_Z).$$

In the following results of [Kaw97, Kaw98, Amb04, Amb05, PS09, FG14], B is a $\mathbb{Q}$-divisor. Kawamata proved that $\mathbf{M}$ is $\boldsymbol{b}$-semi-ample when $\dim X - \dim Z = 1$ and $B \geq 0$ over the generic point of Z [Kaw97] (see [BFMT25, Section 1.2.1]), and that $\mathbf{M}_{Z'}$ is nef on some higher model Z' of Z [Kaw98]. Ambro introduced lc-trivial fibrations, in which $B \geq 0$ is not assumed, under the condition that (X, B) is sub-klt over the generic point of Z [Amb04, Definition 2.1(1)]. When (X, B) is sub-klt over the generic point of Z, $\mathbf{M}$ descends to some higher model Z' of Z and $\mathbf{M}_{Z'}$ is nef [Amb04, Theorem 0.2], and M_Z is semi-ample when $\dim Z = 1$ [Amb04, Theorem 0.1]. Neither result assumes $B \geq 0$. If moreover $B \geq 0$ over the generic point of Z, then there exist a higher model Z' of Z to which $\mathbf{M}$ descends, a contraction $h : Z' \to Z''$, and a nef and big $\mathbb{Q}$-divisor N on Z'' such that

$$\mathbf{M}_{Z'} \sim_{\mathbb{Q}} h^*N$$

Date: November 10, 2026.

2020 *Mathematics Subject Classification.* 14E30, 14D06, 14D07, 14H30.

Key words and phrases. lc-trivial fibration, canonical bundle formula, moduli part, b-semi-ampleness, variation of Hodge structure, cyclic cover, singularities of the base.

[Amb05, Theorem 3.3 and Definition 3.2]. Under the conditions of [PS09, 7.1 and Assumption 7.11], which include $B \geq 0$ over the generic point of Z, Prokhorov and Shokurov conjectured that $\mathbf{M}$ is $\boldsymbol{b}$-semi-ample [PS09, Conjecture 7.13], and proved the conjecture when $\dim X = \dim Z + 1$ using a construction and a result of [Kaw97] [PS09, Theorem 8.1]. Fujino and Gongyo extended the descent and the nefness of $\mathbf{M}$ to the case when (X, B) is only sub-lc over the generic point of Z [FG14, Theorem 3.6] (see [FG14, Definition 3.2(1)]). Recently, [BFMT25] proved the following theorem.

Theorem 1.1 ([BFMT25, Theorem 1.5 and Remark 6.30])**.** *Let $\pi : (X, B) \to Z$ be an lc-trivial fibration such that B is a $\mathbb{Q}$-divisor and $B \geq 0$ over the generic point of Z. Then the moduli $\boldsymbol{b}$-divisor of π is $\boldsymbol{b}$-semi-ample.*

In [BFMT25], lc-trivial fibrations are defined for sub-pairs with coefficients in $\mathbb{Q}$ [BFMT25, Definition 6.18], and [BFMT25, Remark 6.30] shows that the $\boldsymbol{b}$-semi-ampleness of the moduli part of [BFMT25] is equivalent to the $\boldsymbol{b}$-semi-ampleness of $\mathbf{M}$. We ask whether the condition $B \geq 0$ over the generic point of Z can be removed from Theorem 1.1.

Shokurov conjectured that, when the fibers of π are of Fano type, the singularities of (X, B) bound the singularities of the base. In [AB14, Conjecture 1.2] and [Bir16, Conjecture 1.2], as read in [HJL22, Remark 1.6(1)], the conjecture assumes that $B \geq 0$ and (X, B) is ϵ-lc (see Definition 8.1), where X is of Fano type over Z in [AB14] and the general fibers of π are of Fano type in [Bir16]. It predicts that there exists an $\mathbb{R}$-divisor $D \geq 0$ on Z with $D \sim_{\mathbb{R}} M_Z$ such that $(Z, B_Z + D)$ is δ-lc, where $\delta > 0$ depends only on $\dim X$ and ϵ. Han, Jiang, and Luo formulated the conjecture over a point $z \in Z$ in terms of the minimal log discrepancy $\mathrm{mld}(X/Z \ni z, B)$ (see Definition 8.1).

Conjecture 1.2 ([HJL22, Conjecture 1.5])**.** *Let d be a positive integer and ϵ a positive real number. Then there exists a positive real number δ depending only on d and ϵ satisfying the following. Let $\pi : (X, B) \to Z$ be an lc-trivial fibration and $z \in Z$ a (not necessarily closed) point of codimension ≥ 1 such that*

(1) $\dim X - \dim Z = d$,
(2) $\mathrm{mld}(X/Z \ni z, B) \geq \epsilon$, and
(3) the generic fiber of π is of Fano type.

Then there exists an $\mathbb{R}$-divisor $D \geq 0$ on Z with $D \sim_{\mathbb{R}} M_Z$ such that $(Z \ni z, B_Z + D)$ is δ-lc.

Conjecture 1.2 assumes condition (2) in place of the condition in [AB14, Bir16] that $B \geq 0$ and (X, B) is ϵ-lc, so B may be non-effective and (X, B) may have non-klt centers over $Z \backslash \{z\}$ [HJL22, Remark 1.6(1)]. Conjecture 1.2 was known in the following cases:

(1) When $d = 1$ [CH21] (see [HJL22, Remark 1.6(3)]).
(2) When $d = 1$ and $\epsilon \geq 1$, with $\delta = \epsilon - \frac{1}{2}$ [HJL22].

When $B \geq 0$, (X, B) is ϵ-lc, and the couples $(F, \operatorname{Supp} B|_F)$, for the general fibers F of π, belong to a bounded family, the conclusion of Conjecture 1.2 holds with δ depending only on $\dim X$, ϵ, and the bounded family [Bir16]. When $d = 1$, the generic fiber of π is a conic, and the rank condition in Definition 2.4 forces $B \geq 0$ over the generic point of Z (see [HJL22, Remark 2.13]), so in Conjecture 1.2, B can be non-effective over the generic point of Z only when $d \geq 2$.

For $B \geq 0$, Birkar proved a generalized version: if B is a $\mathbb{Q}$-divisor, $B \geq 0$, (X, B) is ϵ-lc, and X is of Fano type over Z, then the generalized pair $(Z, B_Z, \mathbf{M})$ is generalized δ-lc, where $\delta > 0$ depends only on $\dim X - \dim Z$ and ϵ [Bir23, Theorem 1.1]. Birkar and Chen proved a local version for $\mathbb{R}$-divisors: if $T \subset Z$ is a closed subvariety, $B \geq 0$, X is of Fano type over Z, and (X, B) is ϵ-lc along $\pi^{-1}T$, then $(Z, B_Z, \mathbf{M})$ is generalized δ-lc along T, where $\delta > 0$ depends only on $\dim X - \dim Z$ and ϵ [BC25, Theorem 1.3] (see [BC25, Definition 1.1]). When $d = 1$ and $0 < \epsilon \leq 1$, [BC25, Theorem 1.3] was proved earlier with an optimal δ [Che25, Theorem 1.4] (see [BC25, Remark 1.4(1)]).

We ask whether Conjecture 1.2, in which B may be non-effective, holds when $d = 2$.

Main results.

Theorem 1.3. *There exists an lc-trivial fibration $\pi : (X, B) \to Z$ with B a $\mathbb{Q}$-divisor satisfying the following.*

(1) X and Z are smooth projective varieties with $\dim Z = 5$ and $\dim X - \dim Z = 1$.
(2) The generic fiber X_η of π is a smooth curve of genus 1, and $B|_{X_\eta} = \frac{1}{5}D_\eta - P_\eta$ for a $\mathbb{C}(Z)$-rational point P_η of X_η and a reduced divisor D_η on X_η with $D_\eta \sim 5P_\eta$ and $P_\eta \notin \operatorname{Supp} D_\eta$.
(3) The moduli $\boldsymbol{b}$-divisor $\mathbf{M}$ of π descends to Z, and M_Z is nef and not semi-ample.

In particular, $\mathbf{M}$ is not $\boldsymbol{b}$-semi-ample.

By Theorem 1.3(2), B is not ≥ 0 over the generic point of Z, while (X, B) is sub-klt over the generic point of Z. Thus the condition that $B \geq 0$ over the generic point of Z cannot be removed from Theorem 1.1, even when B is a $\mathbb{Q}$-divisor, X and Z are smooth projective, and $\dim X - \dim Z = 1$. The effective form [PS09, (7.13.3)] of the $\boldsymbol{b}$-semi-ampleness conjecture of Prokhorov and Shokurov and its version for real coefficients [Li24, Conjecture 3.5(1)] are equivalent [HLX25, Theorem 1.4]; both assume that the horizontal coefficients of B lie in $[0, 1]$, and Theorem 1.3 shows that this assumption cannot be dropped, even when B is a $\mathbb{Q}$-divisor and (X, B) is sub-klt over the generic point of Z.

For lc-trivial fibrations of relative dimension 2 with generic fiber of Fano type, we obtain the following.

Corollary 1.4. *There exist an lc-trivial fibration $\pi : (X, B) \to Z$ with B a $\mathbb{Q}$-divisor and a closed point $z \in Z$ satisfying the following.*

(1) $\dim X - \dim Z = 2$.
(2) $\operatorname{mld}(X/Z \ni z, B) \geq 1$.
(3) The generic fiber of π is of Fano type.
(4) For any open neighborhood $V \subset Z$ of z and any $\mathbb{R}$-divisor $D \geq 0$ on V with $D \sim_{\mathbb{R}} M_Z|_V$, we have $\operatorname{mld}(V \ni z, B_Z|_V + D) = -\infty$.

In particular, Conjecture 1.2 does not hold for $d = 2$ and $\epsilon = 1$.

Remark 1.5. In Corollary 1.4, we can take $\pi : (X, B) \to Z$ with generic fiber the weighted projective plane $S := \mathbb{P}(1, 2, 3)$ over the function field of Z and with the restriction of B to the generic fiber equal to $E_S + \frac{1}{5}D_S - H$, where H is the unique member of $|\mathcal{O}_S(1)|$, $E_S \in |6H|$ is a smooth curve, and $D_S \in |5H|$ is a curve not containing H (see Section 8). For such π, B is not ≥ 0 over the generic point of Z, and (X, B) is not sub-klt over the generic point of Z. So π does not satisfy the hypotheses of [AB14, Conjecture 1.2] and [Bir16, Conjecture 1.2], as read in [HJL22, Remark 1.6(1)], or of [BC25, Theorem 1.3]. Moreover, we construct such π from an lc-trivial fibration whose generic fiber is S with boundary $E_S + \frac{1}{5}D_S - H$ and whose moduli $\boldsymbol{b}$-divisor is not $\boldsymbol{b}$-semi-ample (see Construction-Proposition 7.1 and Proposition 5.1). By Theorem 1.1, the moduli $\boldsymbol{b}$-divisor of an lc-trivial fibration is $\boldsymbol{b}$-semi-ample when the boundary is a $\mathbb{Q}$-divisor that is ≥ 0 over the generic point of the base.

Sketch of the proof. Let U be the smooth variety of dimension 5 parametrizing the curves E_u of genus 1 with points P_u and reduced divisors $D_u \sim 5P_u$ of Construction-Proposition 4.1. Over a dense open subset $U' \subset Z$, the fibration $\pi : (X, B) \to Z$ of Theorem 1.3 is the base change of the family of $(E_u, \frac{1}{5}D_u - P_u)$ by a finite étale morphism g from U' to an open subset of U (see Section 6). For $(E_u, \frac{1}{5}D_u - P_u)$, the rank condition of Definition 2.4(2) is $h^0(E_u, \mathcal{O}_{E_u}(P_u)) = 1$, since $\lceil \mathbf{A}^*(E_u, \frac{1}{5}D_u - P_u) \rceil = P_u$ (see Definition 2.3 and Section 6), whereas on a conic it forces the boundary to be ≥ 0 (see [HJL22, Remark 2.13]). The canonical bundle formula of [BFMT25, Construction-Definition 6.23 and Theorem 6.28], which does not assume $B \geq 0$, computes $\mathbf{M}$ from the cyclic covers $c_u : C_u \to E_u$ of degree 5 branched along D_u. For the character χ of μ_5 of Construction-Proposition 4.1, the χ-isotypic part of $H^1(C_u, \mathbb{C})$ has Hodge numbers $(1, 4)$,

and its $(1,0)$-parts $H^0(E_u, \omega_{E_u}(P_u))$ form a line bundle $\mathcal{A}$ on U. For the Schmid extension $\mathcal{A}_Z$ [BFMT25, Construction-Definition 6.23] of $g^*\mathcal{A}$, $\mathbf{M}$ descends to Z and

$$\mathcal{O}_Z(mM_Z) \cong \mathcal{A}_Z^{\otimes m}$$

for some $m \in \mathbb{N}^+$ (see Proposition 3.3). Thus Theorem 1.3(3) holds since $\mathcal{A}_Z$ is nef and not semi-ample (see Proposition 5.1). For Corollary 1.4, $\mathcal{A}$ also determines the moduli part of a family of sub-pairs on $\mathbb{P}(1,2,3)$ (see Construction-Proposition 7.1).

Let $\mathbb{V}$ be the variation of Hodge structure on U with fibers

$$\mathbb{V}_u = \operatorname{Ker}\big(c_{u*} : H^1(C_u, \mathbb{Q}) \to H^1(E_u, \mathbb{Q})\big),$$

so that $\mathcal{A} \subset \mathbb{V}_\chi \otimes \mathcal{O}_U$ for the χ-isotypic part $\mathbb{V}_\chi$ of $\mathbb{V} \otimes \mathbb{C}$ (see Construction-Proposition 4.1). After any dominant generically finite base change, the period map of $\mathbb{V}$ is immersive at a general point, and the connected algebraic monodromy group of $\mathbb{V}$ acts irreducibly on $\mathbb{V}_u$ (see Proposition 4.2). By Griffiths' theorem [Voi02, Theorem 10.21] and μ_5-equivariant Serre duality on C_u, the immersivity reduces to the surjectivity of

$$H^0(E_u, \mathcal{O}_{E_u}(2P_u)) \otimes H^0(E_u, \mathcal{O}_{E_u}(3P_u)) \to H^0(E_u, \mathcal{O}_{E_u}(5P_u)),$$

which holds by the base-point-free pencil trick [ACGH85, p. 126], since $H^1(E_u, \mathcal{O}_{E_u}(P_u)) = 0$. For the irreducibility, [Rai25, Theorem 9.2], on μ_N-covers of curves of any genus, implies that the covers of E_u in which one point of D_u moves have large monodromy [Rai25, Definition 3.1], so the connected algebraic monodromy group contains $\operatorname{Res}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}} SU$ of the anti-Hermitian form on $\mathbb{V}_u$ over $\mathbb{Q}(\zeta_5)$ in that definition, and $\operatorname{Res}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}} SU$ acts irreducibly on $\mathbb{V}_u$. The immersivity is seen on $\mathbb{V}_{\chi^2}$, a conjugate of $\mathbb{V}_\chi$ under $\operatorname{Gal}(\mathbb{Q}(\zeta_5)/\mathbb{Q})$ other than $\overline{\mathbb{V}_\chi}$, whose Hodge numbers are $(2,3)$: by the surjectivity above, the map

$$T_{U,u} \to \operatorname{Hom}(\mathbb{V}^{1,0}_{\chi^2,u}, \mathbb{V}^{0,1}_{\chi^2,u}),$$

from a space of dimension 5 to one of dimension 6, given by the Gauss–Manin connection is injective.

Shifting the Hodge types of $\mathbb{V}_\chi$ by $(2,-2)$ and those of $\mathbb{V}_{\chi^{-1}} = \overline{\mathbb{V}_\chi}$ by $(-2,2)$, and keeping those of $\mathbb{V}_{\chi^2}$ and $\mathbb{V}_{\chi^3}$, yields a polarized real variation of Hodge structure $\mathbb{V}'$ with $F^3\mathbb{V}' = \mathcal{A}$ and $F^2\mathbb{V}' = \mathbb{V}_\chi \otimes \mathcal{O}_U$ (compare [BFMT25, Remark 6.26]), since the shifts of $\mathbb{V}_\chi$ and $\overline{\mathbb{V}_\chi}$ are complex conjugate and keep $(\sqrt{-1})^{p-q}$ in the Hodge–Riemann bilinear relations. Thus, by [BBT23, Lemma 6.17], $\mathcal{A}_Z$ is nef, and it is big if and only if the map

$$\psi : T_{U'} \to \operatorname{Hom}\big(g^*\mathcal{A}, (g^*\mathbb{V}_\chi \otimes \mathcal{O}_{U'})/g^*\mathcal{A}\big)$$

given by the Gauss–Manin connection is injective. Since $\operatorname{rank}\psi \le \operatorname{rank}\mathbb{V}_\chi - 1 = 4 < 5 = \dim U'$, ψ is not injective, and $\mathcal{A}_Z$ is not big.

Suppose that $\mathcal{A}_Z$ is semi-ample, and let Z_y be a general fiber of the morphism defined by $|\mathcal{A}_Z^{\otimes n}|$ for some $n \in \mathbb{N}^+$. Then $\dim Z_y \ge 1$ since $\mathcal{A}_Z$ is not big, and ψ vanishes on the tangent spaces of $Z_y \cap U'$ (see (5.1)), so the flat sections of $g^*\mathcal{A}|_{Z_y \cap U'}$ form a rank 1 local subsystem of $g^*\mathbb{V} \otimes \mathbb{C}$ with finite monodromy [Del71, Corollaire (4.2.8)(iii)(b)]. The monodromy group of $g^*\mathbb{V}|_{Z_y \cap U'}$ is normal in that of $g^*\mathbb{V}$, so it is finite (see Proposition 4.2(2)), and the Hodge structure of $g^*\mathbb{V}$ is locally constant along $Z_y \cap U'$, which contradicts Proposition 4.2(1).

The family of Construction-Proposition 7.1 has base W (written Z there) and generic fiber S with boundary $E_S + \frac{1}{5}D_S - H$ as in Remark 1.5. Over a dense open subset of U, a Weierstrass equation embeds E_u into S as $E_S \in |6H|$ with $H|_{E_S} = P_u$, and $D_u = D_S|_{E_S}$ for a unique $D_S \in |5H|$, since the restriction $H^0(S, \mathcal{O}_S(5H)) \to H^0(E_u, \mathcal{O}_{E_u}(5P_u))$ is an isomorphism. Then $K_S + E_S + \frac{1}{5}D_S - H \sim_{\mathbb{Q}} 0$, and by adjunction,

$$\Big(K_S + E_S + \frac{1}{5}D_S - H\Big)\Big|_{E_S} = K_{E_u} + \frac{1}{5}D_u - P_u.$$

The rank condition for $E_S + \frac{1}{5}D_S - H$ is $h^0(S, \mathcal{O}_S(H)) = 1$, which holds since H is the unique member of $|\mathcal{O}_S(1)|$. The cyclic cover of S of degree 5 branched along D_S contains C_u over E_S, and it is a rational surface since the equation of D_S has degree 1 in the coordinate of weight 3 (see (7.1)). Thus, in the χ-isotypic part of H^2 of the complement of the preimage of C_u in a resolution of the cover (see Set-up 3.1), the weight 2 part has $F^2 = 0$, and the residue along C_u is an isomorphism from the weight 3 part onto $H^1(C_u, \mathbb{C})_\chi(-1)$. With the equality in [BFMT25, Construction-Definition 6.23] that passes to the transcendental part, which reduces the Schmid extension of F^2 to the weight 3 part, the canonical bundle formula implies that $\mathbf{M}$ descends to W and $\mathcal{O}_W(mM_W)$ is, for some $m \in \mathbb{N}^+$, the m-th tensor power of the Schmid extension of $g^*\mathcal{A}$ for the morphism $g : U' \to U$ of Construction-Proposition 7.1 (see Proposition 3.3 and Construction-Proposition 7.1(2)). So M_W is nef, not big, and not semi-ample (see Proposition 5.1).

In Section 8, we fix $p \in W$ with $p \in \operatorname{Supp} D$ for any $\mathbb{R}$-divisor $D \geq 0$ with $D \sim_{\mathbb{R}} M_W$, which exists since M_W is not semi-ample, and replace the discriminant of the family by one for which p is a 0-dimensional lc center and $\mathbf{M}$ does not change. Let μ contract the zero section $E \cong W$ of the total space of $\mathcal{O}_W(-A)$, for a very ample divisor A, to the vertex o of the cone Z_0 over (W, A) [Kol13, 3.8]. The fibration $\pi : (X, B) \to Z$ of Corollary 1.4 has base $Z = Z_0 \times \mathbb{A}^1$, and $z = (o, 0)$. For an open neighborhood V of z and an $\mathbb{R}$-divisor $D \geq 0$ on V with $D \sim_{\mathbb{R}} M_Z|_V$, the crepant pullback of $(V, B_Z|_V + D)$ by $\mu \times \mathrm{id}_{\mathbb{A}^1}$ has coefficient $1 + \alpha$ along $E \times \mathbb{A}^1$ for some $\alpha \in \mathbb{R}$, and the strict transform $\widetilde{D}$ of D satisfies

$$\widetilde{D}|_{E\times\{s\}} \sim_{\mathbb{R}} M_W + \alpha A \quad \text{for general } s$$

(see (8.7)). If $\alpha < 0$, then M_W is big, a contradiction. If $\alpha \geq 0$, then $\operatorname{mld}(V \ni z, B_Z|_V + D) = -\infty$, since $1 + \alpha > 1$ when $\alpha > 0$, and $\{p\} \times \mathbb{A}^1 \subset \operatorname{Supp} \widetilde{D}$ is an lc center of the boundary without $\widetilde{D}$ when $\alpha = 0$.

In [BFMT25, Examples 4.6 and 4.7], shifting the Hodge types of two complex conjugate summands as for $\mathbb{V}'$ yields polarized variations whose Hodge bundles are not semi-ample, and [BFMT25, Remark 4.8] says that neither variation is manifestly algebraic. Here the local system $\mathbb{V} \otimes \mathbb{R}$ of $\mathbb{V}'$ is a direct summand of the first cohomology local system of the family of curves C_u.

Structure of the paper. Section 2 recalls lc-trivial fibrations, their discriminants, and their moduli parts. Section 3 shows that, under conditions that hold after shrinking the base, a finite étale base change, and a compactification, the moduli part is $\mathbb{Q}$-linearly equivalent to the Schmid extension of a Hodge line bundle of a family of cyclic covers. Section 4 constructs a family of cyclic covers of degree 5 of curves of genus 1, and studies the period map and the monodromy group of the family. Section 5 proves that the Schmid extension of a Hodge line bundle of the family of Section 4 is nef, not big, and not semi-ample, and Section 6 proves Theorem 1.3. Section 7 constructs a family of sub-pairs on the weighted projective plane $\mathbb{P}(1, 2, 3)$ whose moduli part is $\mathbb{Q}$-linearly equivalent to the Schmid extension of the Hodge line bundle of Section 4, and Section 8 proves Corollary 1.4.

Acknowledgements. The work was partially supported by the National Key R&D Program of China #2024YFA1014400. Jihao Liu would like to thank Bin Dong, Guoxiong Gao, Zeming Sun, Bin Wu, and Leheng Chen for AI assistance. Jihao Liu would like to thank Ruochuan Liu and Gang Tian for consistent support and encouragement.

2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. **Pairs and singularities.**

Definition 2.1. A *pair* $(X/Z \ni z, B)$ consists of a contraction $\pi : X \to Z$ between normal quasi-projective varieties, a point $z \in Z$, and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier over a neighborhood of z. If π is the identity morphism, then we may drop Z and say that $(X \ni z, B)$ is a pair. If $(X/Z \ni z, B)$ is a pair for any $z \in Z$, then we say that $(X/Z, B)$ is a pair and also say that (X, B) is a pair. If we do not require that $B \geq 0$, then we say that $(X/Z \ni z, B)$ is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B).

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

Let $\pi : X \to Z$ be a morphism, (X, B) a sub-pair, and $T \subset Z$ a closed irreducible subset. We say that (X, B) is *sub-lc over* an open subset $V \subset Z$ if $(\pi^{-1}(V), B|_{\pi^{-1}(V)})$ is sub-lc, and *sub-lc over the generic point of* T if (X, B) is sub-lc over some open subset $V \subset Z$ with $V \cap T \neq \emptyset$. We say that $B \geq 0$ *over the generic point of* T if $B|_{\pi^{-1}(V)} \geq 0$ for some open subset $V \subset Z$ with $V \cap T \neq \emptyset$.

2.2. **Lc-trivial fibrations.**

Definition 2.2. Let Z be a normal variety. A ***b**-divisor* $\mathbf{D}$ over Z assigns to each projective birational morphism $Z' \to Z$ from a normal variety an $\mathbb{R}$-divisor $\mathbf{D}_{Z'}$ on Z', the *trace* of $\mathbf{D}$ on Z', such that $g_*\mathbf{D}_{Z''} = \mathbf{D}_{Z'}$ for any projective birational morphism $g : Z'' \to Z'$ over Z. We say that $\mathbf{D}$ *descends* to Z' if $\mathbf{D}_{Z'}$ is $\mathbb{R}$-Cartier and $\mathbf{D}_{Z''} = g^*\mathbf{D}_{Z'}$ for any such g. We say that $\mathbf{D}$ is ***b**-semi-ample* if $\mathbf{D}$ descends to some Z' with $\mathbf{D}_{Z'}$ a semi-ample $\mathbb{Q}$-divisor.

For a ***b**-divisor* $\mathbf{D}$ over Z, $\lceil \mathbf{D} \rceil$ is the ***b**-divisor* with traces $\lceil \mathbf{D}_{Z'} \rceil$, and $\mathcal{O}_Z(\mathbf{D})$ is the sheaf on Z whose sections over an open subset $V \subset Z$ are 0 and the rational functions $\varphi \neq 0$ on Z with $\operatorname{div}(\varphi) + \mathbf{D}_{Z'} \geq 0$ over the preimage of V for any projective birational morphism $Z' \to Z$ from a normal variety.

Definition 2.3. For a sub-pair (X, B), the *discrepancy* ***b**-divisor* $\mathbf{A}(X, B)$ over X has trace

$$\mathbf{A}(X, B)_{X'} := K_{X'} - h^*(K_X + B)$$

on X' for any projective birational morphism $h : X' \to X$ from a normal variety, where $K_{X'}$ is the canonical divisor with $h_*K_{X'} = K_X$, so $\operatorname{mult}_E \mathbf{A}(X, B)_{X'} = a(E, X, B) - 1$ for any prime divisor E on X', and

$$\mathbf{A}^*(X, B)_{X'} := \sum_{\substack{E \subset X' \text{ prime divisor} \\ a(E,X,B)>0}} (a(E, X, B) - 1)E.$$

Definition 2.4 (cf. [HJL22, Definition 2.12] and [BFMT25, Definition 6.18])**.** An *lc-trivial fibration* $\pi : (X, B) \to Z$ consists of a contraction $\pi : X \to Z$ between normal quasi-projective varieties and a sub-pair (X, B) satisfying the following.

(1) (X, B) is sub-lc over the generic point of Z.
(2) $\operatorname{rank} \pi_*\mathcal{O}_X(\lceil \mathbf{A}^*(X, B) \rceil) = 1$.
(3) There exists an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor L on Z with $K_X + B \sim_{\mathbb{R}} \pi^*L$.

Definition 2.5 (cf. [HJL22, (2.1)–(2.2)])**.** Let $\pi : (X, B) \to Z$ be an lc-trivial fibration and L as in Definition 2.4. If B is a $\mathbb{Q}$-divisor, then L is taken to be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor with $K_X + B \sim_{\mathbb{Q}} \pi^*L$, and L, hence M_Z below, is unique up to $\mathbb{Q}$-linear equivalence.

For a prime divisor P on Z, let t_P be the supremum of the real numbers t such that $(X, B+t\pi^*P)$ is sub-lc over the generic point of P, where π^*P is taken over an open subset of Z meeting P on which P is Cartier. The *discriminant* and the *moduli part* of π are

$$B_Z := \sum_P (1-t_P)P \quad \text{and} \quad M_Z := L - K_Z - B_Z.$$

For a projective birational morphism $g : Z' \to Z$ from a normal variety, let X' be a resolution of the main component of $X \times_Z Z'$ with induced morphisms $h : X' \to X$ and $\pi' : X' \to Z'$, and $K_{X'} + B' := h^*(K_X + B)$. Since $\mathbf{A}^*(X', B') = \mathbf{A}^*(X, B)$ and g is an isomorphism over a non-empty open subset of Z, $\pi' : (X', B') \to Z'$ is an lc-trivial fibration. The discriminant and the moduli part of $\pi' : (X', B') \to Z'$, with g^*L in place of L, are the traces on Z' of the *discriminant* $\boldsymbol{b}$*-divisor* $\mathbf{B}$ and the *moduli* $\boldsymbol{b}$*-divisor* $\mathbf{M}$ of π.

The $\boldsymbol{b}$-divisor $\mathbf{A}^*(X, B)$ of [BFMT25, Example 6.1] is $\mathbf{A}(X, B)$ with the coefficients of the prime divisors E with $a(E, X, B) = 0$ set to 0, which coincides with $\mathbf{A}^*(X, B)$ of Definition 2.3 over the preimage of any open subset $V \subset Z$ over which (X, B) is sub-lc, so under Definition 2.4(1), under which a non-empty such V exists, Definition 2.4(2) is [BFMT25, Definition 6.18(ii)]. For a $\mathbb{Q}$-divisor B, the convention on L in Definition 2.5 is the one of [BFMT25, Definition 6.18]. The number $1 - t_P$ is the supremum in [BFMT25, Definition 6.20] (the number $1 + a(E; Y, \Delta)$ in [BFMT25, Definition 6.20] equals $a(E, X, B)$ in our notation), that is, B_Z is the $\mathbb{Q}$-divisor B_X of [BFMT25, Definition 6.20]. The $\mathbb{Q}$-divisor M_Z differs from the Shokurov moduli divisor of [BFMT25, Definition 6.21] only by $\mathbb{Q}$-linear equivalence.

3. The moduli part as a Hodge bundle

Set-up 3.1. Let $\pi : (X, B) \to Z$, Σ, ϕ, and h be as follows.

(1) $\pi : (X, B) \to Z$ is an lc-trivial fibration of relative dimension d between normal projective varieties, Z is smooth, and B is a $\mathbb{Q}$-divisor.
(2) $\Sigma \subset Z$ is an snc divisor such that, for $Z^\circ := Z \setminus \Sigma$, $X^\circ := \pi^{-1}(Z^\circ)$, and $B^\circ := B|_{X^\circ}$, the variety X° is smooth, $\operatorname{Supp} B^\circ$ is an snc divisor, and $(X^\circ, \operatorname{Supp} B^\circ)$ is topologically locally trivial over Z°.
(3) ϕ is a rational function on X such that
$$r(K_{X^\circ} + B^\circ) + \operatorname{div}(\phi)|_{X^\circ} = 0$$
for a canonical divisor K_{X° of X°, where X_η is the generic fiber of π and
$$r := \min\{k \in \mathbb{N}^+ \mid k(K_{X_\eta} + B|_{X_\eta}) \sim 0\}.$$
(4) $f : X' \to X^\circ$ is the normalization of the cyclic cover of X° given by an r-th root $\phi^{1/r}$ of ϕ, with the induced action of μ_r, $(B^\circ)^{=1}$ is the sum of the components of B° with coefficient 1, and $h : X'' \to X'$ is a μ_r-equivariant resolution such that the induced morphism $\pi'' : X'' \to Z^\circ$ is smooth and the reduced inverse image E'' of $(B^\circ)^{=1}$ on X'' is an snc divisor each of whose strata is smooth over Z°.
(5) $R^d\pi''_*\mathbb{Z}_{X'' \setminus E''}$ has unipotent local monodromy around Σ.

Let
$$\mathcal{L} := \mathcal{O}_{X^\circ}(-K_{X^\circ} - \lfloor B^\circ \rfloor),$$
let χ be the character of μ_r with $\zeta \cdot \phi^{1/r} = \chi(\zeta)\phi^{1/r}$ for any $\zeta \in \mu_r$, let $\mathbb{H} := R^d\pi''_*\mathbb{C}_{X'' \setminus E''}$, and let $\mathbb{H}_\chi$ be the χ-isotypic part of $\mathbb{H}$. We denote by $(F^d\mathbb{H}_\chi)_Z$ the Schmid extension [BFMT25, Construction-Definition 6.23] of the deepest Hodge piece $F^d\mathbb{H}_\chi$ of $\mathbb{H}_\chi$.

Conditions (1)–(5) of Set-up 3.1 imply the conditions (†) and (††) of [BFMT25, Construction-Definition 6.23] for the cyclic cover given by an r-th root of ϕ and the resolution h, and $\mathbb{H}_\chi$ is the χ-isotypic part of the variation of mixed Hodge structures $V_Y \otimes \mathbb{C}$ there.

Lemma 3.2 (cf. [BFMT25, Construction-Definition 6.23] and [FG14, Remark 3.9])**.** *Notation as in Set-up 3.1, assuming only conditions (1)–(4) there, and let* $G := (B^\circ)^{=1} - \lfloor B^\circ \rfloor$*. Then the* χ*-isotypic summand of* $f_*\omega_{X'}(f^*(B^\circ)^{=1})$ *is*

$$\omega_{X^\circ}((B^\circ)^{=1}) \otimes \mathcal{L} \cong \mathcal{O}_{X^\circ}(G),$$

and $F^d\mathbb{H}_\chi$ *is a line bundle on* Z° *with*

$$F^d\mathbb{H}_\chi \cong \pi_*\mathcal{O}_{X^\circ}(G) \otimes \omega_{Z^\circ}^{-1}.$$

Proof. Write $B^\circ = \sum_i b_i B_i$, where the B_i are distinct prime divisors, and let $D' := f^*(B^\circ)^{=1}$. Since $(X^\circ, \operatorname{Supp} B^\circ)$ is topologically locally trivial over Z°, any B_i dominates Z°, so $b_i \le 1$ for any i by Definition 2.4(1).

By the minimality of r, $\deg f = r$, and by [EV92, Corollary 3.11],

$$f_*\mathcal{O}_{X'} = \bigoplus_{i=0}^{r-1} \mathcal{L}^{-i}(\lfloor i\{B^\circ\} \rfloor) \cdot \phi^{-i/r},$$

where $\phi^{-i/r} := (\phi^{1/r})^{-i}$ and the i-th summand is χ^{-i}-isotypic. In particular, the χ^{-1}-isotypic summand of $f_*\mathcal{O}_{X'}$ is $\mathcal{L}^{-1} \cdot \phi^{-1/r} \cong \mathcal{L}^{-1}$.

By duality for the finite morphism f, the trace pairing

$$f_*\mathcal{O}_{X'} \otimes f_*\omega_{X'} \to \omega_{X^\circ}$$

induces

$$f_*\omega_{X'} \cong \mathcal{H}om(f_*\mathcal{O}_{X'}, \omega_{X^\circ}).$$

The pairing is μ_r-invariant, so it vanishes on the product of the χ^i-isotypic summand of $f_*\omega_{X'}$ and the χ^{-j}-isotypic summand of $f_*\mathcal{O}_{X'}$ unless $\chi^i = \chi^j$. Thus the χ-isotypic summand of $f_*\omega_{X'}$ is

$$\mathcal{H}om(\mathcal{L}^{-1}, \omega_{X^\circ}) = \omega_{X^\circ} \otimes \mathcal{L}.$$

By Set-up 3.1(3),

$$\operatorname{mult}_{B_i} \operatorname{div}(\phi) = -r \operatorname{mult}_{B_i}(K_{X^\circ} + B^\circ),$$

so the ramification index of f over B_i is the denominator m_i of b_i, and f is étale over the generic point of any prime divisor not contained in $\operatorname{Supp} B^\circ$. In particular, no component of $(B^\circ)^{=1}$ is a branch divisor of f, and D' is reduced. Since $(B^\circ)^{=1}$ is Cartier, the projection formula implies that the χ-isotypic summand of $f_*\omega_{X'}(D')$ is

$$\omega_{X^\circ}((B^\circ)^{=1}) \otimes \mathcal{L} = \mathcal{O}_{X^\circ}(K_{X^\circ} + (B^\circ)^{=1} - K_{X^\circ} - \lfloor B^\circ \rfloor) = \mathcal{O}_{X^\circ}(G).$$

Next we show $h_*\omega_{X''}(E'') = \omega_{X'}(D')$. By the ramification formula, (X', D') is lc and X' is klt, so $h_*\omega_{X''}(E'') = \omega_{X'}(D')$ since E'' contains every h-exceptional prime divisor over $\operatorname{Supp} D'$.

For any $u \in Z^\circ$, we have [Del71, Section 3.2]

$$F^dH^d(X''_u \backslash E''_u, \mathbb{C}) = H^0(X''_u, \omega_{X''_u}(E''_u)).$$

Since π'' is smooth, the strata of E'' are smooth over Z°, and $\dim F^dH^d(X''_u \backslash E''_u, \mathbb{C})$ is independent of u (by the E_1-degeneration of [Del71, Corollaire 3.2.13(ii)], the numbers $h^q(X''_u, \Omega^p_{X''_u}\langle E''_u \rangle)$, $p+q=d$, for the sheaves of logarithmic forms of [Del71, (3.1.2)], are upper semicontinuous in u with constant sum $\dim H^d(X''_u \backslash E''_u, \mathbb{C})$), we have

$$F^d\mathbb{H} = \pi''_*\omega_{X''/Z^\circ}(E'') = \pi''_*\omega_{X''}(E'') \otimes \omega_{Z^\circ}^{-1}.$$

Since μ_r acts trivially on X° and

$$\pi''_*\omega_{X''}(E'') = \pi_*f_*h_*\omega_{X''}(E'') = \pi_*f_*\omega_{X'}(D'),$$

taking χ-isotypic parts implies

$$F^d\mathbb{H}_\chi \cong \pi_*\mathcal{O}_{X^\circ}(G) \otimes \omega_{Z^\circ}^{-1}.$$

Since $F^d\mathbb{H}_\chi$ is locally free, it remains to show $\operatorname{rank}\pi_*\mathcal{O}_{X^\circ}(G)=1$. We show

$$\mathcal{O}_{X^\circ}(\lceil \mathbf{A}^*(X^\circ,B^\circ)\rceil)=\mathcal{O}_{X^\circ}(G),$$

where the left side is the sheaf of Definition 2.2. The coefficient g_i of G along B_i is $-\lfloor b_i\rfloor$ if $b_i<1$ and 0 if $b_i=1$. Thus $0\le g_i<1-b_i$ if $b_i<1$. On X° itself, $a(B_i,X^\circ,B^\circ)=1-b_i$, so

$$\lceil \mathbf{A}^*(X^\circ,B^\circ)_{X^\circ}\rceil=\sum_{b_i<1}\lceil -b_i\rceil B_i=G,$$

which implies $\subset$. Conversely, let $q:Y\to X^\circ$ be a projective birational morphism from a normal variety and E a prime divisor on Y. Since $\operatorname{Supp}B^\circ$ is snc, $a(E,X^\circ,\sum_i B_i)\ge 0$, hence

$$a(E,X^\circ,B^\circ)\ge\sum_i(1-b_i)\operatorname{mult}_E q^*B_i\ge\sum_i g_i\operatorname{mult}_E q^*B_i=\operatorname{mult}_E q^*G,$$

and the second inequality is strict if $\operatorname{mult}_E q^*G>0$. If $\operatorname{mult}_E q^*G>0$, then

$$a(E,X^\circ,B^\circ)>\operatorname{mult}_E q^*G\ge 1,$$

so the coefficient of $\lceil \mathbf{A}^*(X^\circ,B^\circ)_Y\rceil$ along E is $\lceil a(E,X^\circ,B^\circ)-1\rceil\ge\operatorname{mult}_E q^*G$. If $\operatorname{mult}_E q^*G=0$, then the coefficient of $\lceil \mathbf{A}^*(X^\circ,B^\circ)_Y\rceil$ along E is ≥ 0 since $a(E,X^\circ,B^\circ)\ge 0$. Thus $\lceil \mathbf{A}^*(X^\circ,B^\circ)_Y\rceil\ge q^*G$, and any rational function ψ with $\operatorname{div}(\psi)+G\ge 0$ satisfies

$$\operatorname{div}(q^*\psi)+\lceil \mathbf{A}^*(X^\circ,B^\circ)_Y\rceil\ge 0,$$

which implies $\supset$. Restricting Definition 2.4(2) to Z°, we obtain $\operatorname{rank}\pi_*\mathcal{O}_{X^\circ}(G)=1$. □

In [BFMT25, Construction-Definition 6.23], the display for the χ-isotypic summand of the pushforward of logarithmic canonical forms omits the twist by $(B^\circ)^{=1}$, and the divisor there should be $-\lfloor B^{<0}\rfloor=G$, where $B^{<0}$ is the part of B° with negative coefficients. The formula of Lemma 3.2 is the one of [FG14, Remark 3.9].

Proposition 3.3 (cf. [BFMT25, Construction-Definition 6.23, Remark 6.25, and Theorem 6.28])**.** *Notation and conditions as in Set-up 3.1. Then the moduli* $\boldsymbol{b}$*-divisor* $\mathbf{M}$ *of* π *descends to* Z*, and*

$$\mathcal{O}_Z(mM_Z)\cong((F^d\mathbb{H}_\chi)_Z)^{\otimes m}$$

for some positive integer m *such that* mM_Z *is integral.*

Proof. By Lemma 3.2 and the projection formula, the χ-isotypic summand of $f_*\omega_{X'}$ is $\omega_{X^\circ}\otimes\mathcal{L}$, and $F^d\mathbb{H}_\chi$ is a line bundle. Thus χ is the character fixed in [BFMT25, Construction-Definition 6.23] (the one whose isotypic summand of $f_*\omega_{X'}$ is $\omega_{X^\circ}\otimes\mathcal{L}$), and $\mathbb{H}_\chi$ gives the Hodge-theoretic moduli $\boldsymbol{b}$-divisor of π defined there. Let L_Z be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor with $K_X+B\sim_{\mathbb{Q}}\pi^*L_Z$ as in Definition 2.5, so that $M_Z=L_Z-K_Z-B_Z$, and let M^{H} be a divisor on Z with $\mathcal{O}_Z(M^{\mathrm{H}})\cong(F^d\mathbb{H}_\chi)_Z$.

Let $g:Z'\to Z$ be a projective birational morphism from a normal variety Z', possibly $g=\mathrm{id}_Z$, in which case $\mathbf{M}_{Z'}=M_Z$. By [BFMT25, Construction-Definition 6.23 and Remark 6.25], the trace on Z' of the Hodge-theoretic moduli $\boldsymbol{b}$-divisor of π is linearly equivalent to g^*M^{H}. Let $\pi_{Z'}:(X_{Z'},B_{X_{Z'}})\to Z'$ be the lc-trivial fibration over Z' in Definition 2.5, with the induced morphism $h_{Z'}:X_{Z'}\to X$, and let $N_{Z'}$ be its Shokurov moduli divisor [BFMT25, Definition 6.21] formed with $h_{Z'}^*\phi$, that is,

$$r(K_{X_{Z'}}+B_{X_{Z'}})+\operatorname{div}(h_{Z'}^*\phi)=r\pi_{Z'}^*(K_{Z'}+\mathbf{B}_{Z'}+N_{Z'})$$

(the boundary divisor of [BFMT25, Definition 6.20] is the discriminant $\mathbf{B}_{Z'}$). Since $\pi_{Z'}$ is a contraction and $K_{X_{Z'}}+B_{X_{Z'}}\sim_{\mathbb{Q}}\pi_{Z'}^*g^*L_Z$, we have $K_{Z'}+\mathbf{B}_{Z'}+N_{Z'}\sim_{\mathbb{Q}}g^*L_Z$, that is,

$$N_{Z'}\sim_{\mathbb{Q}}g^*L_Z-K_{Z'}-\mathbf{B}_{Z'}=\mathbf{M}_{Z'}.$$

By [BFMT25, Theorem 6.28] for $\pi_{Z'}$, there exists a positive integer c with $cN_{Z'}\sim cg^*M^{\mathrm{H}}$. Thus

$$\mathbf{M}_{Z'}\sim_{\mathbb{Q}}g^*M^{\mathrm{H}}.$$

For $g = \mathrm{id}_Z$, we obtain $M_Z \sim_{\mathbb{Q}} M^{\mathrm{H}}$, so there exists a positive integer m such that mM_Z is integral and $mM_Z \sim mM^{\mathrm{H}}$, which is the second conclusion. For an arbitrary g, we obtain $\mathbf{M}_{Z'} - g^*M_Z \sim_{\mathbb{Q}} 0$, and $g_*(\mathbf{M}_{Z'} - g^*M_Z) = 0$ by Definition 2.2. By the negativity lemma [KM98, Lemma 3.39(1)], $\mathbf{M}_{Z'} = g^*M_Z$. Since Z is smooth, $\mathbf{M}$ descends to Z. □

Lemma 3.4. *Let U be a smooth variety and $\pi_U : X_U \to U$ a smooth projective contraction. Let B_{X_U} be a $\mathbb{Q}$-divisor on X_U such that $\operatorname{Supp} B_{X_U}$ is an snc divisor. Let X_η be the generic fiber of π_U, and let*

$$r := \min\{k \in \mathbb{N}^+ \mid k(K_{X_\eta} + B_{X_U}|_{X_\eta}) \sim 0\}.$$

Let ϕ be a rational function on X_U such that no component of $r(K_{X_U} + B_{X_U}) + \operatorname{div}(\phi)$ dominates U for a canonical divisor K_{X_U} of X_U. Then there exist a dense open subset $U_0 \subset U$, a finite étale morphism $g : U' \to U_0$ from a smooth variety U', and a smooth projective variety Z containing U' as an open subset with the following property. If $\pi : (X, B) \to Z$ is an lc-trivial fibration between normal projective varieties such that B is a $\mathbb{Q}$-divisor and $(\pi^{-1}(U'), B|_{\pi^{-1}(U')})$ is the base change of (X_U, B_{X_U}) by g, then the conditions of Set-up 3.1 hold for π, $\Sigma := Z \setminus U'$, the pullbacks of ϕ and K_{X_U}, and some μ_r-equivariant resolution h.

Proof. Let d be the relative dimension of π_U. We divide the proof into several steps.

Step 1. Since π_U is a smooth contraction, the fibers of π_U are irreducible, so any prime divisor on X_U not dominating U is π_U^*P for a prime divisor P on U. Thus $r(K_{X_U} + B_{X_U}) + \operatorname{div}(\phi) = \pi_U^*Q$ for a $\mathbb{Q}$-divisor Q on U.

Step 2. Let $f_U : X'_U \to X_U$ be the normalization of the cyclic cover of X_U given by an r-th root of ϕ, with the induced action of μ_r, let $h_U : X''_U \to X'_U$ be a μ_r-equivariant log resolution of $(X'_U, f_U^{-1}(\operatorname{Supp} B^{=1}_{X_U}))$, the restriction over U of one given by [AW97, Theorem 0.1] for the normalization of the cyclic cover of a projective compactification of X_U given by an r-th root of ϕ, with the induced μ_r-action, and let E''_U be the reduced inverse image of $B^{=1}_{X_U}$ on X''_U. By generic smoothness and [Ver76, Théorèmes (2.2), (3.3), and (4.14)], since π_U and $X''_U \to U$ are projective, there exists a dense open subset $U_0 \subset U \setminus \operatorname{Supp} Q$ such that the morphism $X''_U \to U$ and each stratum of E''_U are smooth over U_0, and $(X_U, \operatorname{Supp} B_{X_U})$ and (X''_U, E''_U) are topologically locally trivial over U_0. Let X_{U_0}, X'_{U_0}, X''_{U_0}, and E''_{U_0} be the inverse images of U_0 in X_U, X'_U, X''_U, and E''_U, and let $\pi''_{U_0} : X''_{U_0} \to U_0$ be the induced morphism. Then

$$\mathbb{H}_{U_0} := R^d\pi''_{U_0*}\mathbb{Z}_{X''_{U_0} \setminus E''_{U_0}}$$

is a local system on U_0.

Step 3. Fix $u \in U_0$, let Λ be the stalk of $\mathbb{H}_{U_0}$ at u, and let $\Lambda_{\mathrm{tors}} \subset \Lambda$ be the torsion subgroup. Let $g : U' \to U_0$ be the connected finite étale cover corresponding to the kernel of the action of $\pi_1(U_0, u)$ on the finite groups $\Lambda/3\Lambda$ and Λ_{tors}, and let Z be a smooth projective variety containing U' as an open subset such that $Z \setminus U'$ is an snc divisor. Let T be the local monodromy of $g^*\mathbb{H}_{U_0}$ around a component of $Z \setminus U'$. Then T acts trivially on $\Lambda/3\Lambda$ and on Λ_{tors}.

The local system $\mathbb{H}_{U_0} \otimes \mathbb{Q}$ has a filtration W by sub-local systems whose graded pieces are subquotients of the cohomology local systems of X''_{U_0} and of the intersections of components of E''_{U_0} over U_0 [Del71, (3.2.4.1)], and by **Step 2**, X''_{U_0} and the intersections of components of E''_{U_0} are smooth projective over U_0. The local monodromy of the cohomology of a smooth projective family is quasi-unipotent by Borel's lemma [Sch73, 4.5] (see [Kol07, p. 160]), so T is quasi-unipotent. Let ζ be an eigenvalue of T on $\Lambda \otimes \mathbb{C}$. Then ζ is a root of unity. Since T acts trivially on $\Lambda/3\Lambda$, $(T - \mathrm{id}_\Lambda)/3$ is an endomorphism of $\Lambda/\Lambda_{\mathrm{tors}}$, so $(\zeta - 1)/3$ is an algebraic integer. Any conjugate of $(\zeta - 1)/3$ is $(\zeta' - 1)/3$ for a root of unity ζ', hence has absolute value $\leq 2/3 < 1$. If $\zeta \neq 1$, then the norm of the non-zero algebraic integer $(\zeta - 1)/3$ is a non-zero integer of absolute value < 1, which is impossible. Thus $\zeta = 1$, and T is unipotent on $\Lambda/\Lambda_{\mathrm{tors}}$. Since $T - \mathrm{id}_\Lambda$ vanishes on Λ_{tors}, T is unipotent on Λ. The argument of **Step 3** is similar to one used for [BFMT25, Theorem 6.28].

Step 4. Let $\pi : (X, B) \to Z$ be an lc-trivial fibration between normal projective varieties such that B is a $\mathbb{Q}$-divisor and $(\pi^{-1}(U'), B|_{\pi^{-1}(U')})$ is the base change of (X_U, B_{X_U}) by g. Let $X^\circ := \pi^{-1}(U')$, $B^\circ := B|_{X^\circ}$, and let $X_{\eta'}$ be the generic fiber of π. In this step we show

$$\min\{n \in \mathbb{N}^+ \mid n(K_{X_{\eta'}} + B|_{X_{\eta'}}) \sim 0\} = r,$$

and that the base change of h_U by g is a μ_r-equivariant resolution of the normalization of the cyclic cover of X° given by an r-th root of ϕ.

Since π_U is a smooth contraction, X_η is geometrically integral, so by flat base change, for any finite field extension k' of $\mathbb{C}(U)$ and $X_{k'} := X_\eta \otimes_{\mathbb{C}(U)} k'$, a divisor on X_η is linearly equivalent to 0 if and only if its pullback to $X_{k'}$ is. Since $X_{\mathbb{C}(U')} = X_{\eta'}$, this proves the first display of **Step 4**.

We show that $\mathbb{C}(U)$ is algebraically closed in $\mathbb{C}(X_U)(\phi^{1/r})$. Let k' be the algebraic closure of $\mathbb{C}(U)$ in $\mathbb{C}(X_U)(\phi^{1/r})$. Then k' is a finite extension of $\mathbb{C}(U)$. By Kummer theory, the intermediate field $\mathbb{C}(X_U)k'$ of $\mathbb{C}(X_U)(\phi^{1/r})/\mathbb{C}(X_U)$ is $\mathbb{C}(X_U)(\phi^{1/m})$ for a positive integer m dividing r, where $\phi^{1/m} := (\phi^{1/r})^{r/m}$. Since X_η is geometrically integral, $\mathbb{C}(X_U)k'$ is the function field of $X_{k'}$, and by **Step 1**,

$$\frac{r}{m}\big(K_{X_\eta} + B_{X_U}|_{X_\eta}\big) + \operatorname{div}(\phi^{1/m}) = 0$$

on $X_{k'}$. By the previous paragraph, $(r/m)(K_{X_\eta} + B_{X_U}|_{X_\eta}) \sim 0$ on X_η, so $m = 1$ by the minimality of r. Thus $k' \subset \mathbb{C}(X_U)$, and $k' = \mathbb{C}(U)$.

Let X'° and X''° be the base changes of X'_{U_0} and X''_{U_0} by g, and let $h : X''^\circ \to X'^\circ$ be the base change of h_U. Since g is étale and $\mathbb{C}(U)$ is algebraically closed in $\mathbb{C}(X_U)(\phi^{1/r})$, X'° and X''° are normal and irreducible, with function field

$$\mathbb{C}(X_U)(\phi^{1/r}) \otimes_{\mathbb{C}(U)} \mathbb{C}(U') = \mathbb{C}(X^\circ)(\phi^{1/r}).$$

Thus $X'^\circ \to X^\circ$ is the normalization of the cyclic cover of X° given by an r-th root of ϕ, with the induced action of μ_r, and h is a μ_r-equivariant resolution.

Step 5. We verify Set-up 3.1(1)–(5) for π, $\Sigma = Z\backslash U'$, the pullbacks of ϕ and K_{X_U}, and h. Condition (1) holds by assumption and since Z is smooth by **Step 3**. The relative dimension of π is d. By **Step 3**, $\Sigma = Z\backslash U'$ is an snc divisor. Since $X^\circ \to X_{U_0}$ is étale, X° is smooth, $\operatorname{Supp} B^\circ$ is an snc divisor, and the pullback of K_{X_U} is a canonical divisor of X°. By **Step 2**, $(X^\circ, \operatorname{Supp} B^\circ)$ is topologically locally trivial over U'. This is (2). Condition (3) follows from **Step 4** and from the equality of **Step 1** over $U_0 \subset U \setminus \operatorname{Supp} Q$, pulled back by the étale morphism $X^\circ \to X_{U_0}$. By **Steps 2 and 4**, h satisfies (4): the induced morphism $X''^\circ \to U'$ is smooth, and the reduced inverse image of $(B^\circ)^{=1}$ on X''° is the base change of E''_{U_0}, an snc divisor each of whose strata is smooth over U'. Finally, for the induced morphism $\pi'' : X''^\circ \to U'$ and the reduced inverse image E'' of $(B^\circ)^{=1}$, we have

$$R^d\pi''_*\mathbb{Z}_{X''^\circ\setminus E''} = g^*\mathbb{H}_{U_0},$$

so (5) follows from **Step 3**. □

Lemma 3.5. *Let $\pi : X \to Z$ be a contraction between normal projective varieties with X $\mathbb{Q}$-factorial, B^h a $\mathbb{Q}$-divisor on X, r a positive integer, and ϕ a rational function on X such that no component of $r(K_X + B^h) + \operatorname{div}(\phi)$ dominates Z for a canonical divisor K_X of X, and let $B^v := -\frac{1}{r}(r(K_X + B^h) + \operatorname{div}(\phi))$. Assume that (X, B^h) is sub-lc over the generic point of Z and $\operatorname{rank} \pi_*\mathcal{O}_X(\lceil \mathbf{A}^*(X, B^h)\rceil) = 1$. Then $\pi : (X, B^h + B^v) \to Z$ is an lc-trivial fibration with $K_X + B^h + B^v \sim_{\mathbb{Q}} 0$.*

Proof. By the definition of B^v, $r(K_X + B^h + B^v) + \operatorname{div}(\phi) = 0$, and $(X, B^h + B^v)$ coincides with (X, B^h) over a non-empty open subset of Z. □

4. Period map and monodromy of cyclic covers of elliptic curves

In this section, we construct a family $\mathcal{C} \to U$ of cyclic covers of degree 5 of curves of genus 1 over a smooth base U of dimension 5, and we study the period map and the monodromy group of $\mathcal{C} \to U$ (Proposition 4.2).

Construction-Proposition 4.1. Let χ be the character $\zeta \to \zeta^{-1}$ of μ_5. Then there exist a smooth variety U of dimension 5, a smooth projective morphism $e : \mathcal{E} \to U$ with a section $P \subset \mathcal{E}$, a reduced divisor $D \subset \mathcal{E}$ finite and étale of degree 5 over U with $D \cap P = \emptyset$, and a μ_5-cover $c : \mathcal{C} \to \mathcal{E}$ with $e \circ c$ smooth and projective, satisfying the following, where E_u, P_u, D_u, and C_u are the fibers over $u \in U$.

(1) E_u is a curve of genus 1, $D_u \sim 5P_u$, and $c_u : C_u \to E_u$ is the cyclic cover of degree 5 branched along D_u with $c_{u*}\mathcal{O}_{C_u} \cong \bigoplus_{i=0}^{4} \mathcal{O}_{E_u}(-iP_u)$, where μ_5 acts on $\mathcal{O}_{E_u}(-iP_u)$ by χ^{-i}.

(2) For any $u \in U$, the Kodaira–Spencer map $T_{U,u} \to H^1(E_u, T_{E_u}(-D_u))$ of the family $(\mathcal{E}, D) \to U$ at u is an isomorphism.

(3) The kernel $\mathbb{V}$ of the trace map

$$R^1(e \circ c)_*\mathbb{Q}_{\mathcal{C}} \to R^1 e_* \mathbb{Q}_{\mathcal{E}}$$

is a polarized variation of $\mathbb{Q}$-Hodge structure of weight 1 and rank 20 with an action of $\mathbb{Q}(\zeta_5)$, where ζ_5 is a primitive fifth root of unity, and

$$\mathbb{V} \otimes \mathbb{C} = \bigoplus_{i=1}^{4} \mathbb{V}_{\chi^i}$$

for the χ^i-isotypic parts $\mathbb{V}_{\chi^i}$ of $\mathbb{V} \otimes \mathbb{C}$, with $\operatorname{rank} \mathbb{V}_{\chi^i} = 5$ and $\mathbb{V}^{1,0}_{\chi^i,u} = H^0(E_u, \omega_{E_u}(iP_u))$, where

$$\mathbb{V}^{p,q}_{\chi^i,u} := \mathbb{V}^{p,q}_u \cap \mathbb{V}_{\chi^i,u}$$

and $H^0(E_u, \omega_{E_u}(iP_u))$ is embedded into $H^0(C_u, \omega_{C_u})$ by $\eta \to c_u^*\eta \cdot (\xi^{1/5}|_{C_u})^{-i}$ for the rational function $\xi^{1/5}$ on $\mathcal{C}$ in (5).

(4) $\mathcal{A} := \mathbb{V}^{1,0}_{\chi}$ is a line subbundle of $\mathbb{V}_{\chi} \otimes \mathcal{O}_U$.

(5) c is the normalization of $\mathcal{E}$ in $\mathbb{C}(\mathcal{E})(\xi^{1/5})$ for a rational function ξ on $\mathcal{E}$ with $\operatorname{div}(\xi) = D - 5P$, and μ_5 acts on $\xi^{1/5}$ by χ^{-1}.

Proof. Let $\mathcal{M}$ be the moduli stack of pairs (E, D), where E is a smooth projective curve of genus 1 and $D \geq 0$ is a reduced divisor of degree 5 on E. The tangent space of $\mathcal{M}$ at (E, D) is $H^1(E, T_E(-D))$ [ACG11, Chapter XI, Section 2 and Theorem (2.12)]. Since $\deg T_E(-D) = -5$, by the Riemann–Roch theorem,

$$h^0(E, T_E(-D)) = 0 \quad \text{and} \quad h^1(E, T_E(-D)) = 5.$$

Since $h^0(E, T_E(-D)) = 0$ and the obstruction space $H^2(E, T_E(-D))$ vanishes, $\mathcal{M}$ is a smooth Deligne–Mumford stack with $\dim \mathcal{M} = h^1(E, T_E(-D)) = 5$.

Let $\mathcal{R}$ be the moduli stack of quadruples (E, L, D, α), where (E, D) is a pair parametrized by $\mathcal{M}$, L is a line bundle of degree 1 on E, and $\alpha : L^{\otimes 5} \to \mathcal{O}_E(D)$ is an isomorphism. Isomorphisms of quadruples include isomorphisms $E \to E'$ of the curves. The fiber of $\mathcal{R} \to \mathcal{M}$ over (E, D) is $\{L \mid L^{\otimes 5} \cong \mathcal{O}_E(D)\}$, a torsor under $\operatorname{Pic}^0(E)[5]$, and α is unique up to the isomorphisms $\lambda^{1/5} \operatorname{id}_L$ for $\lambda \in \mathbb{C}^*$. Thus $\mathcal{R} \to \mathcal{M}$ is étale. By the Riemann–Roch theorem, $L \cong \mathcal{O}_E(P)$ for a unique point $P \in E$, and $D \in |5P| \cong \mathbb{P}^4$. Since $|5P|$ is base point free, the divisors in $|5P|$ containing P form a hyperplane of $|5P|$. Let $\mathcal{Z} \subset \mathcal{R}$ be the locus where $P \in \operatorname{Supp} D$. Then

$$\dim \mathcal{Z} \leq \dim \mathcal{M}_{1,1} + \dim\{D \in |5P| \mid P \in \operatorname{Supp} D\} = 1 + 3 = 4,$$

where $\mathcal{M}_{1,1}$ is the moduli stack of curves of genus 1 with a point.

To (E, L, D, α) we attach the cyclic cover

$$c_E : C := \operatorname{Spec}_E \bigoplus_{i=0}^{4} L^{\otimes -i} \to E,$$

where, for $i + j \geq 5$, the multiplication is the composition

$$L^{\otimes -i} \otimes L^{\otimes -j} \to L^{\otimes -(i+j)} \cong L^{\otimes -(i+j-5)}(-D) \subset L^{\otimes -(i+j-5)},$$

the isomorphism being induced by α, and $\zeta \in \mu_5$ acts on $L^{\otimes -i}$ by ζ^i [EV92, 3.5 and 3.9]. Thus μ_5 acts on $L^{\otimes -i}$ by χ^{-i}. Near a point of D with local coordinate x on E with $D = (x = 0)$, the curve C is given by $y^5 = x$, since D is reduced. Thus C is smooth, c_E is totally ramified over D and étale elsewhere, and C is connected, because any connected component of C maps onto E and the fiber of c_E over a point of D is one point. Since $\deg D = 5 \deg L = 5$, by the Riemann–Hurwitz formula,

$$2g(C) - 2 = 5(2g(E) - 2) + 4 \deg D = 20, \quad \text{so} \quad g(C) = 11.$$

A level-3 structure on (E, L, D, α) is an isomorphism $H^1(C, \mathbb{Z}/3) \cong (\mathbb{Z}/3)^{22}$. An automorphism of (E, L, D, α) induces an automorphism of C, and the induced automorphism determines the automorphism of the quadruple, since it determines the automorphism of $E = C/\mu_5$ and of the summand $L^{\otimes -1}$ of $c_{E*}\mathcal{O}_C$. Since $g(C) = 11 \geq 2$, an automorphism of C acting trivially on $H^1(C, \mathbb{Z}/3)$ is the identity (Serre, [ACG11, Chapter XVI, Proposition (2.8)]). Thus the moduli stack $\mathcal{R}^{(3)}$ of quadruples with a level-3 structure has trivial automorphism groups, so $\mathcal{R}^{(3)}$ is an algebraic space [Stacks, Tag 04SZ], finite and étale over $\mathcal{R}$, and hence étale over $\mathcal{M}$. Let U_0 be a connected component of a dense open subscheme [Stacks, Tag 06NH] of $\mathcal{R}^{(3)}$. It is a smooth variety with $\dim U_0 = \dim \mathcal{M} = 5$. Let $U \subset U_0$ be the open subset of points u such that $P_u \notin \operatorname{Supp} D_u$, where $L_u \cong \mathcal{O}_{E_u}(P_u)$ for the quadruple $(E_u, L_u, D_u, \alpha_u)$ at u. Since $U_0 \to \mathcal{R}$ is quasi-finite and $\dim \mathcal{Z} \leq 4$, U is dense in U_0.

Let $(\mathcal{E}, \mathcal{L}, D, \alpha)$ be the universal quadruple over U, $e : \mathcal{E} \to U$ the family of curves, and

$$c : \mathcal{C} := \operatorname{Spec}_{\mathcal{E}} \bigoplus_{i=0}^{4} \mathcal{L}^{\otimes -i} \to \mathcal{E}$$

the cyclic cover attached to $(\mathcal{E}, \mathcal{L}, D, \alpha)$, whose multiplication and μ_5-action are those of c_E with $\mathcal{L}$ in place of L. Since $h^0(E_u, \mathcal{L}|_{E_u}) = 1$ and $h^1(E_u, \mathcal{L}|_{E_u}) = 0$ for any $u \in U$, $e_*\mathcal{L}$ is a line bundle. Possibly shrinking U, we may assume that $e_*\mathcal{L} \cong \mathcal{O}_U$. A generator σ of $e_*\mathcal{L}$ is a section of $\mathcal{L}$ whose zero divisor P satisfies $P \cap E_u = P_u$ for any $u \in U$. Thus P is a section of e, $\mathcal{L} \cong \mathcal{O}_{\mathcal{E}}(P)$ via σ, and $D \cap P = \emptyset$ by the definition of U. The divisor D is finite and flat of degree 5 over U with reduced fibers, so D is étale over U. Near D, the variety $\mathcal{C}$ is given by $y^5 = x$ for a local equation x of D, and c is étale over $\mathcal{E} \backslash D$. Since D is étale over U, $\mathcal{C}$ is smooth over U, so $e \circ c$ is smooth and projective.

(1) The restriction of c to E_u is the cover c_{E_u} attached to $(E_u, \mathcal{L}|_{E_u}, D_u, \alpha|_{E_u})$, and

$$\mathcal{L}|_{E_u} \cong \mathcal{O}_{E_u}(P_u).$$

Thus (1) follows from the properties of c_E above, and $D_u \sim 5P_u$ via α.

(2) The Kodaira–Spencer map of $(\mathcal{E}, D)$ at u is the tangent map at u of the classifying morphism $U \to \mathcal{M}$. The classifying morphism is the composition of the open immersion $U \to \mathcal{R}^{(3)}$ and the étale morphisms $\mathcal{R}^{(3)} \to \mathcal{R} \to \mathcal{M}$, so the tangent map is an isomorphism.

(5) Via σ, α is an isomorphism $\mathcal{O}_{\mathcal{E}}(5P) \to \mathcal{O}_{\mathcal{E}}(D)$ of subsheaves of the sheaf of rational functions on $\mathcal{E}$, so α is the multiplication by ξ^{-1} for a rational function ξ on $\mathcal{E}$ with $\operatorname{div}(\xi) = D - 5P$. Since $D \neq 0$ is reduced, ξ is not the fifth power of a rational function on $\mathcal{E}$, so $[\mathbb{C}(\mathcal{E})(\xi^{1/5}) : \mathbb{C}(\mathcal{E})] = 5$. The map

$$\bigoplus_{i=0}^{4} \mathcal{O}_{\mathcal{E}}(-iP) \to \mathbb{C}(\mathcal{E})(\xi^{1/5}), \quad (s_i)_{i=0}^4 \to \sum_{i=0}^{4} s_i \xi^{i/5},$$

is injective, and it is a homomorphism of $\mathcal{O}_{\mathcal{E}}$-algebras, since $s\xi^{i/5}\cdot s'\xi^{j/5}=(ss'\xi)\cdot\xi^{(i+j-5)/5}$ for $i+j\geq 5$. Thus $\mathcal{C}$ is finite over $\mathcal{E}$ with function field $\mathbb{C}(\mathcal{E})(\xi^{1/5})$. Since $\mathcal{C}$ is smooth, $\mathcal{C}$ is the normalization of $\mathcal{E}$ in $\mathbb{C}(\mathcal{E})(\xi^{1/5})$. The summand $\mathcal{O}_{\mathcal{E}}(-P)$, on which μ_5 acts by χ^{-1}, is $\mathcal{O}_{\mathcal{E}}(-P)\cdot\xi^{1/5}$, so μ_5 acts on $\xi^{1/5}$ by χ^{-1}.

(3) Since $e\circ c$ is smooth and projective, $R^1(e\circ c)_*\mathbb{Q}_{\mathcal{C}}$ is a polarized variation of $\mathbb{Q}$-Hodge structure of weight 1, and the trace map is a morphism of variations, so $\mathbb{V}$ is a polarized sub-variation. For any $u\in U$, the trace c_{u*} and the pullback c_u^* satisfy

$$c_{u*}\circ c_u^*=5\quad\text{and}\quad c_u^*\circ c_{u*}=\sum_{\zeta\in\mu_5}\zeta^*,$$

so $\mathbb{V}_u$ is the kernel of $\sum_{\zeta\in\mu_5}\zeta^*$ on $H^1(C_u,\mathbb{Q})$. Thus

$$\mathbb{V}\otimes\mathbb{C}=\bigoplus_{i=1}^{4}\mathbb{V}_{\chi^i},$$

and the group algebra $\mathbb{Q}[\mu_5]$ acts on $\mathbb{V}$ through

$$\mathbb{Q}[\mu_5]/\Big(\sum_{\zeta\in\mu_5}\zeta\Big)\cong\mathbb{Q}(\zeta_5).$$

Let $u\in U$, $\xi_u:=\xi|_{E_u}$, and $\xi_u^{-i/5}:=(\xi^{1/5}|_{C_u})^{-i}$ for $0\leq i\leq 4$. Then $\operatorname{div}(\xi_u)=D_u-5P_u$. Let $\widetilde{D}_u:=c_u^{-1}(D_u)_{\mathrm{red}}$, so $c_u^*D_u=5\widetilde{D}_u$ and the ramification divisor of c_u is $4\widetilde{D}_u$. For $0\leq i\leq 4$ and $\eta\in H^0(E_u,\omega_{E_u}(iP_u))$, we have

$$\operatorname{div}(c_u^*\eta\cdot\xi_u^{-i/5})=c_u^*(\operatorname{div}(\eta)+iP_u)+(4-i)\widetilde{D}_u\geq 0,$$

and μ_5 acts on $\xi_u^{-i/5}$ by χ^i. Thus

$$H^0(E_u,\omega_{E_u}(iP_u))\to H^0(C_u,\omega_{C_u})_{\chi^i},\quad \eta\to c_u^*\eta\cdot\xi_u^{-i/5},$$

is an injective linear map for $0\leq i\leq 4$. Since

$$\sum_{i=0}^{4}h^0(E_u,\omega_{E_u}(iP_u))=1+1+2+3+4=11=g(C_u)=h^0(C_u,\omega_{C_u}),$$

all five maps are isomorphisms. Thus, for $1\leq i\leq 4$,

$$\mathbb{V}^{1,0}_{\chi^i,u}=H^0(C_u,\omega_{C_u})_{\chi^i}\cong H^0(E_u,\omega_{E_u}(iP_u))\quad\text{and}\quad\dim\mathbb{V}^{1,0}_{\chi^i,u}=i.$$

Since μ_5 acts on $\mathbb{V}_u$ by $\mathbb{Q}$-linear maps, complex conjugation maps $\mathbb{V}_{\chi^i,u}$ onto $\mathbb{V}_{\chi^{-i},u}$, so

$$\dim\mathbb{V}^{0,1}_{\chi^i,u}=\dim\mathbb{V}^{1,0}_{\chi^{5-i},u}=5-i.$$

Thus $\operatorname{rank}\mathbb{V}_{\chi^i}=5$ for $1\leq i\leq 4$ and $\operatorname{rank}\mathbb{V}=20$.

(4) By (3), $\dim\mathbb{V}^{1,0}_{\chi,u}=1$ for any $u\in U$. Since $\mathbb{V}^{1,0}$ is a holomorphic subbundle of $\mathbb{V}\otimes\mathcal{O}_U$ stable under μ_5, its image $\mathbb{V}^{1,0}_{\chi}$ under the projector $\frac{1}{5}\sum_{\zeta\in\mu_5}\chi(\zeta)^{-1}\zeta^*$ onto $\mathbb{V}_\chi\otimes\mathcal{O}_U$ is a line subbundle of $\mathbb{V}_\chi\otimes\mathcal{O}_U$. □

Proposition 4.2. *Notation as in Construction-Proposition 4.1. Let $g:U'\to U$ be a dominant generically finite morphism from a smooth variety U'. Then the following hold.*

(1) For a general point $u'\in U'$, the map

$$T_{U',u'}\to\operatorname{Hom}(\mathbb{V}^{1,0}_{g(u')},\mathbb{V}^{0,1}_{g(u')})$$

given by the Gauss–Manin connection of $g^*\mathbb{V}$ is injective.

(2) For any $u'\in U'$, the identity component of the Zariski closure in $GL(\mathbb{V}_{g(u')})$ of the monodromy group of $g^*\mathbb{V}$ acts irreducibly on the $\mathbb{Q}$-vector space $\mathbb{V}_{g(u')}$.

Proof. (1) Since g is dominant and generically finite, $\dim U' = \dim U = 5$, and g is étale at a general point $u' \in U'$ by generic smoothness. At such u', the map in (1) is the composition of the isomorphism $dg : T_{U',u'} \to T_{U,g(u')}$ and the map

$$\overline{\nabla}_u : T_{U,u} \to \mathrm{Hom}(\mathbb{V}_u^{1,0}, \mathbb{V}_u^{0,1})$$

given by the Gauss–Manin connection of $\mathbb{V}$ at $u = g(u')$. Thus it suffices to prove that $\overline{\nabla}_u$ is injective for any $u \in U$.

Fix $u \in U$ and write E, P, D, $\widetilde{D}$, C, c, ξ for E_u, P_u, D_u, $\widetilde{D}_u$, C_u, c_u, $\xi|_{E_u}$, and $\xi^{-j/5}$ for $\xi_u^{-j/5}$. Let

$$\kappa : T_{U,u} \to H^1(E, T_E(-D))$$

be the Kodaira–Spencer map of $(\mathcal{E}, D)$, an isomorphism by Construction-Proposition 4.1(2), and let $\kappa_{\mathcal{C}} : T_{U,u} \to H^1(C, T_C)$ be the Kodaira–Spencer map of $e \circ c$. By Griffiths' theorem [Voi02, Theorem 10.21], for $v \in T_{U,u}$, $\overline{\nabla}_u(v)$ is the restriction to the subspace $\mathbb{V}_u^{1,0}$ of $H^0(C, \omega_C)$ of the cup product

$$\kappa_{\mathcal{C}}(v) \cup - : H^0(C, \omega_C) \to H^1(C, \mathcal{O}_C).$$

Near a point of D, the curve C is given by $y^5 = x$ for a local coordinate x on E with $D = (x = 0)$, and the μ_5-invariant vector fields on C are the multiples of $y\partial_y = 5x\partial_x$ by invariant functions. Away from D, c is étale. Thus $(c_* T_C)^{\mu_5} = T_E(-D)$, and since c is finite and taking μ_5-invariants is exact,

$$\iota : H^1(E, T_E(-D)) = H^1(E, (c_* T_C)^{\mu_5}) \xrightarrow{\cong} H^1(C, T_C)^{\mu_5}.$$

By the local description of c above, a vector field on an open subset of E vanishing on D lifts uniquely to a μ_5-invariant vector field on its preimage in C. Thus the first-order deformation of C along v is obtained by gluing with the lifts of the gluing vector fields of the first-order deformation of (E, D), and $\kappa_{\mathcal{C}} = \iota \circ \kappa$.

Let $\mathrm{tr}_C : H^1(C, \omega_C) \to \mathbb{C}$ be the trace map. For $\alpha \in H^0(C, \omega_C)_{\chi^2}$ and $\beta \in H^0(C, \omega_C)_{\chi^3}$, we have

$$\mathrm{tr}_C\left((\kappa_{\mathcal{C}}(v) \cup \alpha) \cup \beta\right) = \mathrm{tr}_C\left(\kappa_{\mathcal{C}}(v) \cup (\alpha\beta)\right),$$

where $\alpha\beta \in H^0(C, \omega_C^{\otimes 2})^{\mu_5}$ since $\chi^2\chi^3 = 1$. Since the Serre duality pairing between $H^1(C, T_C)$ and $H^0(C, \omega_C^{\otimes 2})$ is μ_5-invariant, it restricts to a perfect pairing between $H^1(C, T_C)^{\mu_5}$ and $H^0(C, \omega_C^{\otimes 2})^{\mu_5}$. Let $v \in \mathrm{Ker}\, \overline{\nabla}_u$. Since $\mathbb{V}^{1,0}_{\chi^2,u} = H^0(C, \omega_C)_{\chi^2}$, we have $\kappa_{\mathcal{C}}(v) \cup \alpha = 0$ for any $\alpha \in H^0(C, \omega_C)_{\chi^2}$. Thus, if

$$H^0(C, \omega_C^{\otimes 2})^{\mu_5} = \mathrm{Span}\{\alpha\beta \mid \alpha \in H^0(C, \omega_C)_{\chi^2},\ \beta \in H^0(C, \omega_C)_{\chi^3}\}, \tag{4.1}$$

then $\kappa_{\mathcal{C}}(v) = 0$, so $v = 0$ since $\iota \circ \kappa$ is injective (compare [LO11, Proposition 4.1], for base curves of genus ≥ 2).

We prove (4.1). By Construction-Proposition 4.1(3), any

$$\alpha \in H^0(C, \omega_C)_{\chi^2} \quad \text{and} \quad \beta \in H^0(C, \omega_C)_{\chi^3}$$

are $\alpha = c^*\eta_2 \cdot \xi^{-2/5}$ and $\beta = c^*\eta_3 \cdot \xi^{-3/5}$ with $\eta_j \in H^0(E, \omega_E(jP))$, so

$$\alpha\beta = c^*(\eta_2\eta_3\xi^{-1}), \quad \text{where} \quad \eta_2\eta_3\xi^{-1} \in H^0(E, \omega_E^{\otimes 2}(D))$$

since $\mathrm{div}(\xi^{-1}) = 5P - D$. For $q \in H^0(E, \omega_E^{\otimes 2}(D))$,

$$\mathrm{div}(c^*q) = c^*(\mathrm{div}(q) + D) + 3\widetilde{D} \geq 0,$$

so the map

$$c^* : H^0(E, \omega_E^{\otimes 2}(D)) \to H^0(C, \omega_C^{\otimes 2})^{\mu_5}$$

is injective. By the Riemann–Roch theorem and Serre duality,

$$h^0(E, \omega_E^{\otimes 2}(D)) = 5 = h^1(E, T_E(-D)) = \dim H^1(C, T_C)^{\mu_5} = \dim H^0(C, \omega_C^{\otimes 2})^{\mu_5},$$

so c^* is an isomorphism. Let $L := \mathcal{O}_E(P)$. Since $\omega_E \cong \mathcal{O}_E$ and multiplication by ξ^{-1} is an isomorphism

$$H^0(E, \omega_E^{\otimes 2}(5P)) \to H^0(E, \omega_E^{\otimes 2}(D)),$$

(4.1) follows from the surjectivity of the multiplication map

$$H^0(E, L^{\otimes 2}) \otimes H^0(E, L^{\otimes 3}) \to H^0(E, L^{\otimes 5}).$$

Since $L^{\otimes 2}$ is base point free with $h^0(E, L^{\otimes 2}) = 2$ and $H^1(E, L) = 0$, the multiplication map is surjective by the base-point-free pencil trick [ACGH85, p. 126]. This proves (1).

(2) Let $u \in U$, $V := \mathbb{V}_u$, $\Gamma \subset GL(V)$ the monodromy group of $\mathbb{V}$ at u, that is, the image of $\pi_1(U, u)$, and $\overline{\Gamma}$ the Zariski closure of Γ in $GL(V)$. We call the identity component $\overline{\Gamma}^0$ the *connected algebraic monodromy group* of $\mathbb{V}$ at u. We first prove that $\overline{\Gamma}^0$ acts irreducibly on V.

We show that $\overline{\Gamma}^0$ contains the group G defined below. Write $D_u = x_1 + \cdots + x_5$. By [Rai25, Theorem 9.2] (on μ_N-covers of a curve of genus g with $n + 2 - 2g$ ramification points for $n \geq \max\{5, 2g - 1\}$), applied with $N = 5$, $g = 1$, and $n = 5$ in the notation there, the family of μ_5-covers of E_u containing c_u in which the branch point x_1 moves in $E_u \backslash \{x_2, \ldots, x_5\}$ and $x_2, \ldots, x_5$ are fixed has large monodromy in the sense of [Rai25, Definition 3.1]. Let b_0 be a point of the base T_0 of the family in [Rai25, Theorem 9.2] over which the cover is c_u, and let T be the connected component of T_0 containing b_0, a smooth curve finite étale over a dense open subset of E_u (see the beginning of [Rai25, Section 9]). Since 5 is prime, the intermediate covers of c_u are C_u and E_u, so the Prym of [Rai25, (1.3)] is isogenous to the kernel of the trace. Thus the Prym local system on T has fiber V at b_0, with the lattice $V_{\mathbb{Z}} := V \cap H^1(C_u, \mathbb{Z})$, which is preserved by the monodromy, and we let $\Gamma_T \subset GL(V)$ be the monodromy group at b_0. Let $\langle \cdot, \cdot \rangle$ be the non-degenerate anti-Hermitian form on the $\mathbb{Q}(\zeta_5)$-vector space V, with respect to $\mathbb{Q}(\zeta_5)/\mathbb{Q}(\sqrt{5})$, preserved by Γ_T in [Rai25, Definition 3.1], and let

$$G := \mathrm{Res}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}\, SU(V, \langle \cdot, \cdot \rangle) \subset GL(V),$$

a connected algebraic group over $\mathbb{Q}$. Let S be the set of finite places $\mathfrak{p}$ of $\mathbb{Q}(\sqrt{5})$ of [Rai25, Definition 3.1]. Only finitely many finite places are not in S. Since $V_{\mathbb{Z}}$ and the Γ_T-stable lattice of [Rai25, Definition 3.1] are commensurable, removing finitely many places from S, we may assume that their completions at $\mathfrak{p}$ coincide for any $\mathfrak{p} \in S$. For $\mathfrak{p} \in S$, let $SU(V_{\mathbb{Z},\mathfrak{p}})$ be the stabilizer of the completion $V_{\mathbb{Z},\mathfrak{p}}$ of $V_{\mathbb{Z}}$ at $\mathfrak{p}$ in the group

$$SU(V, \langle \cdot, \cdot \rangle)(\mathbb{Q}(\sqrt{5})_{\mathfrak{p}}).$$

Largeness means that $\Gamma_T^1 := \Gamma_T \cap G(\mathbb{Q})$ has dense image in $\prod_{\mathfrak{p} \in S} SU(V_{\mathbb{Z},\mathfrak{p}})$.

Let ℓ be a prime number that splits in $\mathbb{Q}(\sqrt{5})$ as

$$\ell \mathcal{O}_{\mathbb{Q}(\sqrt{5})} = \mathfrak{p}_1 \mathfrak{p}_2 \quad \text{with} \quad \mathfrak{p}_1, \mathfrak{p}_2 \in S.$$

Such ℓ exists, since there exist infinitely many primes $\ell \equiv \pm 1 \bmod 5$, each of them splits in $\mathbb{Q}(\sqrt{5})$, and only finitely many finite places are not in S. Then

$$\mathbb{Q}(\sqrt{5}) \otimes \mathbb{Q}_\ell = \mathbb{Q}(\sqrt{5})_{\mathfrak{p}_1} \times \mathbb{Q}(\sqrt{5})_{\mathfrak{p}_2},$$

so the group

$$SU(V_{\mathbb{Z},\mathfrak{p}_1}) \times SU(V_{\mathbb{Z},\mathfrak{p}_2})$$

is an open subgroup of $G(\mathbb{Q}_\ell)$. It contains Γ_T^1 as a dense subgroup, since Γ_T^1 preserves $V_{\mathbb{Z}}$ and has dense image in $\prod_{\mathfrak{p} \in S} SU(V_{\mathbb{Z},\mathfrak{p}})$. Let $\overline{\Gamma_T^1}$ be the Zariski closure of Γ_T^1 in $GL(V)$. Since $\overline{\Gamma_T^1}(\mathbb{Q}_\ell)$ is closed in $GL(V \otimes \mathbb{Q}_\ell)$ for the ℓ-adic topology, $(\overline{\Gamma_T^1} \cap G)(\mathbb{Q}_\ell)$ contains an open subgroup of $G(\mathbb{Q}_\ell)$. Thus

$$\mathrm{Lie}(\overline{\Gamma_T^1} \cap G) \otimes \mathbb{Q}_\ell = \mathrm{Lie}(G) \otimes \mathbb{Q}_\ell,$$

so $\dim(\overline{\Gamma_T^1} \cap G) = \dim G$, and $G \subset (\overline{\Gamma_T^1})^0$ since G is connected. Thus $G \subset \overline{\Gamma_T}^0$.

Let $T' \to T$ be a connected finite étale cover over which the $\mathrm{Pic}^0(E_u)[5]$-torsor of the roots L and the $GL_{22}(\mathbb{Z}/3)$-torsor of level-3 structures are trivial, let $b_0' \in T'$ be a point over b_0, and let $T'^\circ \subset T'$ be a dense open subset containing b_0' over which an isomorphism α exists. Then the pullback of the family to T'° is the cyclic cover attached to a family of quadruples with a level-3 structure over T'°, equal at b_0' to the quadruple and the level-3 structure of u. Since T'° is connected, the classifying morphism $T'^\circ \to \mathcal{R}^{(3)}$ maps T'° into the connected component of $\mathcal{R}^{(3)}$ containing u. Since U is open in $\mathcal{R}^{(3)}$ and contains u, possibly removing finitely many points from T'°, we may assume that it is a morphism $T'^\circ \to U$. The pullback of $\mathbb{V}$ by $T'^\circ \to U$ is the Prym local system, so the image $\Gamma_{T'}$ of $\pi_1(T'^\circ, b_0')$ in $GL(V)$ is contained in Γ. Since $\pi_1(T'^\circ, b_0') \to \pi_1(T', b_0')$ is surjective [Kol95, Proposition 2.10] and the image of $\pi_1(T', b_0')$ in $\pi_1(T, b_0)$ has finite index, $\Gamma_{T'}$ has finite index in Γ_T, so

$$G \subset \overline{\Gamma_T}^0 = \overline{\Gamma_{T'}}^0 \subset \overline{\Gamma}^0.$$

We show that G acts irreducibly on V, which implies that $\overline{\Gamma}^0$ acts irreducibly on V. Let $\tau_1, \overline{\tau}_1, \tau_2, \overline{\tau}_2$ be the embeddings of $\mathbb{Q}(\zeta_5)$ into $\overline{\mathbb{Q}}$, where

$$\tau_1|_{\mathbb{Q}(\sqrt{5})} \neq \tau_2|_{\mathbb{Q}(\sqrt{5})},$$

where $\overline{\tau}_j := \tau_j \circ \gamma$ for the complex conjugation γ of $\mathbb{Q}(\zeta_5)$, and let $V_\tau := V \otimes_{\mathbb{Q}(\zeta_5),\tau} \overline{\mathbb{Q}}$, so

$$V \otimes_{\mathbb{Q}} \overline{\mathbb{Q}} = V_{\tau_1} \oplus V_{\overline{\tau}_1} \oplus V_{\tau_2} \oplus V_{\overline{\tau}_2} \quad \text{and} \quad \dim V_\tau = 5.$$

Then

$$G_{\overline{\mathbb{Q}}} \cong SL(V_{\tau_1}) \times SL(V_{\tau_2}),$$

where $SL(V_{\tau_j})$ acts trivially on $V_{\tau_{3-j}}$ and $V_{\overline{\tau}_{3-j}}$, and acts on $V_{\overline{\tau}_j}$ as the dual of V_{τ_j} via the perfect pairing $V_{\tau_j} \times V_{\overline{\tau}_j} \to \overline{\mathbb{Q}}$ induced by $\langle \cdot, \cdot \rangle$. Since the standard representation of SL_5 is irreducible and not isomorphic to the dual representation, the four summands are pairwise non-isomorphic irreducible representations of $G_{\overline{\mathbb{Q}}}$. Let $W \subset V$ be a G-invariant $\mathbb{Q}$-subspace. Then $W \otimes \overline{\mathbb{Q}}$ is a sum of some of the four summands, and it is stable under $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, which permutes the four summands transitively since it acts transitively on $\{\tau_1, \overline{\tau}_1, \tau_2, \overline{\tau}_2\}$. Thus $W = 0$ or $W = V$.

Now let $u' \in U'$. Since g is dominant and generically finite and U' is irreducible, there exists a dense open subset $U_1 \subset U$ such that $g^{-1}(U_1) \to U_1$ is finite and étale. Since U' is connected, parallel transport along a path from u' to a point of $g^{-1}(U_1)$ conjugates the monodromy groups of $g^*\mathbb{V}$ and their Zariski closures, so we may assume that $u' \in g^{-1}(U_1)$. Let $u := g(u')$, and let Γ be the monodromy group of $\mathbb{V}$ at u. The image of $\pi_1(g^{-1}(U_1), u')$ in $\pi_1(U_1, u)$ has finite index, and the maps

$$\pi_1(U_1, u) \to \pi_1(U, u) \quad \text{and} \quad \pi_1(g^{-1}(U_1), u') \to \pi_1(U', u')$$

are surjective [Kol95, Proposition 2.10]. Thus the monodromy group of $g^*\mathbb{V}$ at u' is a finite index subgroup of Γ, so the connected algebraic monodromy group of $g^*\mathbb{V}$ at u' is $\overline{\Gamma}^0$, which acts irreducibly on $V = \mathbb{V}_{g(u')}$. □

5. A nef Hodge line bundle that is not semi-ample

Proposition 5.1. *Notation as in Construction-Proposition 4.1. Let $g : U' \to U$ be a dominant generically finite morphism from a smooth variety U', and let Z be a smooth projective variety containing U' as an open subset. Assume that $Z \backslash U'$ is an snc divisor and $g^*\mathbb{V}$ has unipotent local monodromy around $Z \backslash U'$. Let $j : U' \to Z$ be the inclusion, let $\overline{\mathcal{V}}$ be the Deligne canonical extension of $g^*\mathbb{V} \otimes \mathcal{O}_{U'}$ [Del70, Chapter II, Proposition 5.2], and let $\mathcal{A}_Z := j_* g^* \mathcal{A} \cap \overline{\mathcal{V}}$ be the Schmid extension of $g^*\mathcal{A}$. Then $\mathcal{A}_Z$ is nef, not big, and not semi-ample.*

Proof of Proposition 5.1. We divide the proof into two steps. Let ∇ be the Gauss–Manin connection of $\mathbb{V} \otimes \mathcal{O}_U$. We also write ∇ for the Gauss–Manin connection of any pullback of $\mathbb{V}$. Since ∇ preserves $g^*\mathbb{V}_\chi \otimes \mathcal{O}_{U'}$, we may let

$$\psi : T_{U'} \to \mathrm{Hom}\big(g^*\mathcal{A}, (g^*\mathbb{V}_\chi \otimes \mathcal{O}_{U'})/g^*\mathcal{A}\big)$$

be the $\mathcal{O}_{U'}$-linear map sending a local vector field v to the map $s \to \nabla_v s \bmod g^*\mathcal{A}$. For $x \in U'$, let ψ_x be the fiber of ψ at x.

Step 1. In this step we prove that $\mathcal{A}_Z$ is nef and not big.

Since $\chi^5 = 1$, we have $\mathbb{V}_{\chi^{-1}} = \mathbb{V}_{\chi^4}$. For any $u \in U$ and $i \in \{1,2,3,4\}$, by Construction-Proposition 4.1(3),

$$\mathbb{V}_{\chi^i,u} = \mathbb{V}^{1,0}_{\chi^i,u} \oplus \mathbb{V}^{0,1}_{\chi^i,u} \quad \text{and} \quad \overline{\mathbb{V}^{p,q}_{\chi^i,u}} = \mathbb{V}^{q,p}_{\chi^{-i},u} \text{ for any } (p,q) \in \{(1,0),(0,1)\}.$$

We set $\mathbb{V}^{p,q}_{\chi^i,u} := 0$ for any other $(p,q) \in \mathbb{Z}^2$ with $p+q=1$. For any $(p,q) \in \mathbb{Z}^2$ with $p+q=1$, let

$$\mathbb{V}'^{p,q}_{\chi,u} := \mathbb{V}^{p-2,q+2}_{\chi,u}, \quad \mathbb{V}'^{p,q}_{\chi^{-1},u} := \mathbb{V}^{p+2,q-2}_{\chi^{-1},u}, \quad \text{and} \quad \mathbb{V}'^{p,q}_{\chi^i,u} := \mathbb{V}^{p,q}_{\chi^i,u} \text{ for } i \in \{2,3\}$$

(compare [BFMT25, Remark 6.26]), and let

$$\mathbb{V}'^{p,q}_u := \bigoplus_{i=1}^{4} \mathbb{V}'^{p,q}_{\chi^i,u}.$$

Then

$$\mathbb{V}_u \otimes \mathbb{C} = \bigoplus_{p\in\mathbb{Z}} \mathbb{V}'^{p,1-p}_u \quad \text{and} \quad \overline{\mathbb{V}'^{p,q}_u} = \mathbb{V}'^{q,p}_u.$$

For $k \in \mathbb{Z}$, let $F^k\mathbb{V}_{\chi^i}$ be $\mathbb{V}_{\chi^i} \otimes \mathcal{O}_U$ if $k \le 0$, $\mathbb{V}^{1,0}_{\chi^i}$ if $k=1$, and 0 if $k \ge 2$. For $p \in \mathbb{Z}$, let

$$F^p\mathbb{V}' := F^{p-2}\mathbb{V}_\chi \oplus F^p\mathbb{V}_{\chi^2} \oplus F^p\mathbb{V}_{\chi^3} \oplus F^{p+2}\mathbb{V}_{\chi^{-1}},$$

a holomorphic subbundle of $\mathbb{V} \otimes \mathcal{O}_U$ with fiber $\bigoplus_{k\ge p} \mathbb{V}'^{k,1-k}_u$ at $u \in U$. Then

$$F^4\mathbb{V}' = 0, \qquad F^3\mathbb{V}' = \mathbb{V}^{1,0}_\chi = \mathcal{A}, \qquad \text{and} \qquad F^2\mathbb{V}' = \mathbb{V}_\chi \otimes \mathcal{O}_U.$$

Since ∇ preserves $\mathbb{V}_{\chi^i} \otimes \mathcal{O}_U$ and

$$\nabla F^k\mathbb{V}_{\chi^i} \subset F^{k-1}\mathbb{V}_{\chi^i} \otimes \Omega^1_U$$

for any i and k, we have

$$\nabla F^p\mathbb{V}' \subset F^{p-1}\mathbb{V}' \otimes \Omega^1_U$$

for any p.

Let Q be the polarization of $\mathbb{V}$ given by the cup product on $H^1(C_u,\mathbb{Q})$. Since μ_5 acts on C_u by automorphisms, Q is μ_5-invariant. Thus $Q(\mathbb{V}_{\chi^i,u},\mathbb{V}_{\chi^j,u}) = 0$ unless $i+j=5$, and the subspaces $\mathbb{V}_{\chi^i,u}$, $i \in \{1,2,3,4\}$, are pairwise orthogonal for the sesquilinear form $(v,w) \to Q(v,\overline{w})$. For any i and (p,q), $\mathbb{V}'^{p,q}_{\chi^i,u} = \mathbb{V}^{p',q'}_{\chi^i,u}$ with $(p'-q')-(p-q) \in \{-4,0,4\}$, so $(\sqrt{-1})^{p-q} = (\sqrt{-1})^{p'-q'}$. Since Q satisfies the Hodge–Riemann bilinear relations for the decomposition $\bigoplus_{p,q} \mathbb{V}^{p,q}_{\chi^i,u}$, it satisfies them for the decomposition $\bigoplus_{p,q} \mathbb{V}'^{p,q}_u$. Therefore

$$\mathbb{V}' := (\mathbb{V} \otimes \mathbb{R}, F^\bullet\mathbb{V}', Q)$$

is a polarized real variation of Hodge structure of weight 1 on U.

The pullback $g^*\mathbb{V}'$ is a polarized real variation of Hodge structure on U' with

$$F^3g^*\mathbb{V}' = g^*\mathcal{A} \quad \text{and} \quad F^2g^*\mathbb{V}' = g^*\mathbb{V}_\chi \otimes \mathcal{O}_{U'},$$

so ψ is the map $T_{U'} \to \operatorname{Hom}(F^3, F^2/F^3)$ of [BBT23, Lemma 6.17] for $g^*\mathbb{V}'$. Since the underlying local system of $g^*\mathbb{V}'$ is $g^*\mathbb{V} \otimes \mathbb{R}$, $g^*\mathbb{V}'$ has unipotent local monodromy around $Z\backslash U'$, the Deligne canonical extension of $g^*\mathbb{V}'$ on Z is $\overline{\mathcal{V}}$, and the Schmid extension of $F^3g^*\mathbb{V}'$ is $\mathcal{A}_Z$. Since g is dominant and generically finite,

$$\operatorname{rank} T_{U'} = \dim U = 5 > 4 = \operatorname{rank} \operatorname{Hom}\big(g^*\mathcal{A}, (g^*\mathbb{V}_\chi \otimes \mathcal{O}_{U'})/g^*\mathcal{A}\big),$$

so ψ is not injective. By [BBT23, Lemma 6.17], $\det(\mathcal{A}_Z)$ is nef, and it is big if and only if ψ is injective as a map of $\mathcal{O}_{U'}$-modules. Thus $\mathcal{A}_Z = \det(\mathcal{A}_Z)$ is nef and not big.

Step 2. In this step we prove that $\mathcal{A}_Z$ is not semi-ample.

Suppose that $\mathcal{A}_Z$ is semi-ample. Let m be a positive integer such that $|\mathcal{A}_Z^{\otimes m}|$ is base point free, and let $Z \xrightarrow{f} Y \to \mathbb{P}^N$ be the Stein factorization of the morphism defined by $|\mathcal{A}_Z^{\otimes m}|$. Then f is a contraction onto a normal projective variety Y, and $\mathcal{A}_Z^{\otimes m} \cong f^*\mathcal{H}$ for an ample line bundle $\mathcal{H}$ on Y. Since $\mathcal{A}_Z$ is not big, $\dim Y < \dim Z = 5$. Let $y \in Y$ be a general point, let $Z_y := f^{-1}(y)$, and let $Z_y^\circ := Z_y \cap U'$. Then Z_y is a smooth connected projective variety, Z_y° is a non-empty open subset of Z_y, and

$$\dim Z_y = 5 - \dim Y \geq 1, \qquad \mathcal{A}_Z^{\otimes m}|_{Z_y} \cong f^*\mathcal{H}|_{Z_y} \cong \mathcal{O}_{Z_y}.$$

We show that

$$\psi_x(v) = 0 \quad \text{for any } x \in Z_y^\circ \text{ and any } v \in T_x Z_y. \tag{5.1}$$

We may assume that $v \neq 0$. Let $n := \dim Z_y$, let $\mathcal{L}$ be a very ample line bundle on Z_y, and let $H_1, \dots, H_{n-1}$ be general members of $|\mathcal{L}|$ with $x \in H_j$ and $v \in T_x H_j$ for any j. Since $\mathcal{L}$ is very ample, $H_1 \cap \cdots \cap H_{n-1}$ is smooth of dimension 1 at x with tangent line $\mathbb{C}v$. Let Λ be the irreducible component of $H_1 \cap \cdots \cap H_{n-1}$ containing x (so $\Lambda = Z_y$ if $n = 1$), let $\nu : \widetilde{\Lambda} \to Z$ be the composition of the normalization of Λ with the inclusion $\Lambda \subset Z$, and let $\widetilde{\Lambda}^\circ := \nu^{-1}(U')$. Since $x \in \Lambda \cap U'$, $\widetilde{\Lambda} \backslash \widetilde{\Lambda}^\circ$ is a finite set. For any $t \in \widetilde{\Lambda} \backslash \widetilde{\Lambda}^\circ$, ν maps a small loop around t to a product of powers of small loops around the components of $Z \backslash U'$ through $\nu(t)$. The local monodromies of $g^*\mathbb{V}$ around the components of $Z \backslash U'$ through $\nu(t)$ commute and are unipotent, so $\nu^* g^* \mathbb{V}$ has unipotent local monodromy around t. By [BFMT25, Lemma 2.16(3)], the Schmid extension of $F^3 \nu^* g^* \mathbb{V}' = \nu^* g^* \mathcal{A}$ on $\widetilde{\Lambda}$ is $\nu^* \mathcal{A}_Z$. Since $\nu(\widetilde{\Lambda}) \subset Z_y$,

$$m \deg \nu^* \mathcal{A}_Z = \deg \nu^* \big(\mathcal{A}_Z^{\otimes m}|_{Z_y} \big) = 0,$$

so $\nu^* \mathcal{A}_Z$ is not big. Let

$$\psi_{\widetilde{\Lambda}} : T_{\widetilde{\Lambda}^\circ} \to \operatorname{Hom}\big(\nu^* g^* \mathcal{A}, (\nu^* g^* \mathbb{V}_\chi \otimes \mathcal{O}_{\widetilde{\Lambda}^\circ}) / \nu^* g^* \mathcal{A} \big)$$

be the $\mathcal{O}_{\widetilde{\Lambda}^\circ}$-linear map sending a local vector field ∂ to the map $s \to \nabla_\partial s \bmod \nu^* g^* \mathcal{A}$. Then $\psi_{\widetilde{\Lambda}} = \nu^* \psi \circ d\nu$. By [BBT23, Lemma 6.17], $\psi_{\widetilde{\Lambda}}$ is not injective. Since $T_{\widetilde{\Lambda}^\circ}$ is a line bundle on the integral curve $\widetilde{\Lambda}^\circ$, $\psi_{\widetilde{\Lambda}} = 0$. Since Λ is smooth at x, $\nu^{-1}(x)$ is a single point $\widetilde{x}$ and $d\nu(T_{\widetilde{x}} \widetilde{\Lambda}) = T_x \Lambda = \mathbb{C}v$. Thus $\psi_x(v) = 0$, which proves (5.1).

By (5.1), $\nabla_v s$ is a section of $g^*\mathcal{A}$ for any local section s of $g^*\mathcal{A}|_{Z_y^\circ}$ and any local vector field v on Z_y°. Thus the flat connection of $g^*\mathbb{V} \otimes \mathcal{O}_{Z_y^\circ}$ preserves the line subbundle $g^*\mathcal{A}|_{Z_y^\circ}$, and the flat sections of $g^*\mathcal{A}|_{Z_y^\circ}$ form a rank 1 local subsystem $\mathbb{W} \subset g^*\mathbb{V}_\chi|_{Z_y^\circ}$ with $\mathbb{W} \otimes \mathcal{O}_{Z_y^\circ} = g^*\mathcal{A}|_{Z_y^\circ}$. Since Z_y is smooth and connected, Z_y° is connected. Let $e_y : \mathcal{C}_y \to Z_y^\circ$ be the base change of the smooth projective morphism $e \circ c$ along $g|_{Z_y^\circ}$. Since the composition of the pullback

$$R^1 e_* \mathbb{Q}_\mathcal{E} \to R^1 (e \circ c)_* \mathbb{Q}_\mathcal{C}$$

with the trace map is multiplication by $\deg c = 5$,

$$R^1 (e \circ c)_* \mathbb{Q}_\mathcal{C} \cong R^1 e_* \mathbb{Q}_\mathcal{E} \oplus \mathbb{V}.$$

Thus $g^*\mathbb{V}|_{Z_y^\circ}$ is a direct summand of $R^1 e_{y*} \mathbb{Q}_{\mathcal{C}_y}$ that is pointwise a sub-Hodge structure (Construction-Proposition 4.1(3)), and $\mathbb{W}$ is a rank 1 complex local subsystem of $R^1 e_{y*} \mathbb{C}_{\mathcal{C}_y}$. By [Del71, Corollaire (4.2.8)(iii)(b)], a positive tensor power of $\mathbb{W}$ is a trivial local system. Fix a point $x \in Z_y^\circ$. Then the kernel of the monodromy representation $\pi_1(Z_y^\circ, x) \to GL(\mathbb{W}_x)$ is a subgroup of finite index, and it fixes a non-zero vector $w \in \mathbb{W}_x \subset \mathbb{V}_{g(x)} \otimes \mathbb{C}$.

By [Ver76, Corollaire 5.1], there exists a dense open subset $Y^\circ \subset Y$ over which $f|_{U'} : U' \to Y$ is a topologically locally trivial fibration. Since y is general, $y \in Y^\circ$. Let $U'^\circ := U' \cap f^{-1}(Y^\circ)$. Then $U'^\circ \to Y^\circ$ is a topologically locally trivial fibration with fiber Z_y° over y, and by the homotopy exact sequence, the image of $\pi_1(Z_y^\circ, x)$ in $\pi_1(U'^\circ, x)$ is the kernel of $\pi_1(U'^\circ, x) \to \pi_1(Y^\circ, y)$, hence a normal subgroup. Let $\rho : \pi_1(U', x) \to GL(\mathbb{V}_{g(x)})$ be the monodromy representation of $g^*\mathbb{V}$, let $\Pi := \rho(\pi_1(U', x))$, and let $\Pi_y \subset \Pi$ be the image of $\pi_1(Z_y^\circ, x)$. Since U' is smooth and $U'^\circ \subset U'$ is

a dense open subset, $\pi_1(U'^\circ, x) \to \pi_1(U', x)$ is surjective [Kol95, Proposition 2.10]. Thus Π is the image of $\pi_1(U'^\circ, x)$, so Π_y is a normal subgroup of Π.

Let $\overline{\Pi}$ and $\overline{\Pi_y}$ be the Zariski closures of Π and Π_y in $GL(\mathbb{V}_{g(x)})$, and let $\overline{\Pi}^0$ and $\overline{\Pi_y}^0$ be their identity components. Since Π_y is normal in Π, $\overline{\Pi_y}^0$ is normal in $\overline{\Pi}$, and in particular $\overline{\Pi}^0$ normalizes $\overline{\Pi_y}^0$.

Since the image in Π_y of the kernel of $\pi_1(Z_y^\circ, x) \to GL(\mathbb{W}_x)$ has finite index and fixes w, its Zariski closure contains $\overline{\Pi_y}^0$, so $\overline{\Pi_y}^0$ fixes w. Let $V_0 \subset \mathbb{V}_{g(x)}$ be the $\mathbb{Q}$-subspace of vectors fixed by $\overline{\Pi_y}^0$. Since $\overline{\Pi_y}^0$ is defined over $\mathbb{Q}$, the subspace of $\mathbb{V}_{g(x)} \otimes \mathbb{C}$ fixed by $\overline{\Pi_y}^0$ is $V_0 \otimes \mathbb{C}$, which contains w. Thus $V_0 \neq 0$. Since $\overline{\Pi}^0$ normalizes $\overline{\Pi_y}^0$, $\overline{\Pi}^0$ preserves V_0. By Proposition 4.2(2), $V_0 = \mathbb{V}_{g(x)}$. Thus $\overline{\Pi_y}^0$ is trivial, and Π_y is finite.

For any $x' \in U'$, let

$$\overline{\nabla}_{x'} : T_{U',x'} \to \mathrm{Hom}(\mathbb{V}^{1,0}_{g(x')}, \mathbb{V}^{0,1}_{g(x')})$$

be the map of Proposition 4.2(1). The monodromy group of $g^*\mathbb{V}|_{Z_y^\circ}$ is Π_y. Since Π_y is finite, by [Del71, (4.1.3.3)] the Hodge structure of $g^*\mathbb{V}|_{Z_y^\circ}$ is locally constant, that is, $\nabla_v s$ is a section of $g^*\mathbb{V}^{1,0}$ for any local section s of $g^*\mathbb{V}^{1,0}|_{Z_y^\circ}$ and any local vector field v on Z_y°. Thus $\overline{\nabla}_{x'}$ vanishes on $T_{x'}Z_y$ for any $x' \in Z_y^\circ$. By Proposition 4.2(1), there exists a dense open subset $U_1' \subset U'$ such that $\overline{\nabla}_{x'}$ is injective for any $x' \in U_1'$. Since $f(U_1')$ contains a dense open subset of Y and y is general, there exists $x' \in Z_y \cap U_1' \subset Z_y^\circ$. Since $\dim Z_y \geq 1$, $T_{x'}Z_y \neq 0$, which contradicts the injectivity at x'. Thus $\mathcal{A}_Z$ is not semi-ample. □

6. Proof of Theorem 1.3

Proof of Theorem 1.3. Let U, $e : \mathcal{E} \to U$, P, D, $c : \mathcal{C} \to \mathcal{E}$, χ, $\mathbb{V}$, $\mathcal{A}$, and ξ be as in Construction-Proposition 4.1, and let $\phi := \xi^{-1}$. Since E_u is a curve of genus 1 for any $u \in U$, possibly shrinking U, we may assume that $\omega_{\mathcal{E}} \cong \mathcal{O}_{\mathcal{E}}$, and we take $K_{\mathcal{E}} := 0$. By Construction-Proposition 4.1(5), $\mathrm{div}(\phi) = 5P - D$, so

$$5\Big(K_{\mathcal{E}} + \frac{1}{5}D - P\Big) + \mathrm{div}(\phi) = 0.$$

Since $D \neq 0$ is reduced, $k(\frac{1}{5}D - P)$ is integral only if $5 \mid k$. Thus the integer r of Lemma 3.4 for $(\mathcal{E}, \frac{1}{5}D - P)$ is 5.

By Lemma 3.4 for $\pi_U := e$, $B_{X_U} := \frac{1}{5}D - P$, and ϕ, there exist a dense open subset $U_0 \subset U$, a finite étale morphism $g : U' \to U_0$ from a smooth variety U', and a smooth projective variety Z containing U' as an open subset with the property in Lemma 3.4. Fix an embedding $\mathcal{E} \times_U U' \subset \mathbb{P}^N \times U'$ over U'. A resolution of the closure of $\mathcal{E} \times_U U'$ in $\mathbb{P}^N \times Z$ that is an isomorphism over U' is a smooth projective variety X with a contraction $\pi : X \to Z$ such that $\pi^{-1}(U') = \mathcal{E} \times_U U'$. We let

- D_X and P_X be the closures in X of the pullbacks of D and P, and $B^h := \frac{1}{5}D_X - P_X$,
- ϕ_X be the pullback of ϕ to X, and
- K_X be a canonical divisor of X with $K_X|_{\pi^{-1}(U')} = 0$.

Then $(5(K_X + B^h) + \mathrm{div}(\phi_X))|_{\pi^{-1}(U')} = 0$, so no component of $5(K_X + B^h) + \mathrm{div}(\phi_X)$ dominates Z, and $B^v := -\frac{1}{5}(5(K_X + B^h) + \mathrm{div}(\phi_X))$ satisfies $B^v|_{\pi^{-1}(U')} = 0$.

Let X_η be the generic fiber of π, and $D_\eta := D_X|_{X_\eta}$ and $P_\eta := P_X|_{X_\eta}$. Then the following hold.

(i) X_η is a smooth curve of genus 1 and D_η is reduced.
(ii) P_η is a point of degree 1 on X_η, since P_X is the closure of a section of π over U'.
(iii) $P_\eta \notin \mathrm{Supp}\, D_\eta$, since $D \cap P = \emptyset$.
(iv) $D_\eta - 5P_\eta = -\mathrm{div}(\phi_X)|_{X_\eta}$, so $D_\eta \sim 5P_\eta$.

For any exceptional prime divisor T over X, $\dim \operatorname{center}_X T \le \dim X - 2 = \dim Z - 1$, so any prime divisor T over X whose center dominates Z is a horizontal prime divisor on X, and

$$a(T, X, B^h) = \begin{cases} 2 & \text{if } T = P_X, \\ \frac{4}{5} & \text{if } T \subset D_X, \\ 1 & \text{otherwise.} \end{cases}$$

Over U', $\operatorname{Supp} B^h$ is snc with coefficients $\frac{1}{5}$ and -1, so (X, B^h) is sub-klt over U'. Moreover, $\lceil \mathbf{A}^*(X, B^h) \rceil$ restricts to P_η on X_η by the display, so

$$\operatorname{rank} \pi_* \mathcal{O}_X(\lceil \mathbf{A}^*(X, B^h) \rceil) = h^0(X_\eta, \mathcal{O}_{X_\eta}(P_\eta)) = 1.$$

By Lemma 3.5, $\pi : (X, B) \to Z$ is an lc-trivial fibration for the $\mathbb{Q}$-divisor $B := B^h + B^v$, and $(\pi^{-1}(U'), B|_{\pi^{-1}(U')})$ is the base change of $(\mathcal{E}, \frac{1}{5}D - P)$ by g. Since $B|_{X_\eta} = \frac{1}{5}D_\eta - P_\eta$ and $\dim Z = \dim U' = \dim U = 5$, (1) and (2) hold.

By the property of Z in Lemma 3.4, the conditions of Set-up 3.1 hold for π, $\Sigma := Z \backslash U'$, ϕ_X, the pullback of $K_{\mathcal{E}}$, and a μ_5-equivariant resolution $X'' \to X'$, where X' is the normalization of X° in $\mathbb{C}(X)(\phi_X^{1/5})$ as in Set-up 3.1(4), with $d = 1$, $r = 5$, $X^\circ = \mathcal{E} \times_U U'$, and $(B^\circ)^{=1} = 0$, so $E'' = \emptyset$. Since $\phi^{1/5} = (\xi^{1/5})^{-1}$ and $e \circ c$ is smooth, Construction-Proposition 4.1(5) implies $X' = \mathcal{C} \times_U U'$. For a general point $u' \in U'$, the resolution $X'' \to X'$ restricts to a birational morphism $X''_{u'} \to C_{g(u')}$ between smooth projective curves, which is an isomorphism. Thus the μ_5-equivariant morphism of local systems

$$g^* R^1 (e \circ c)_* \mathbb{C}_{\mathcal{C}} \to \mathbb{H}$$

induced by $X'' \to X'$ is an isomorphism on U'. By Construction-Proposition 4.1(5), $\zeta \cdot \phi_X^{1/5} = \chi(\zeta)\phi_X^{1/5}$ for any $\zeta \in \mu_5$, so the character χ of Set-up 3.1 is the character χ of Construction-Proposition 4.1. Since χ is non-trivial and the μ_5-invariant part of $H^1(C_u, \mathbb{C})$ is the pullback of $H^1(E_u, \mathbb{C})$, the χ-isotypic part of $g^* R^1 (e \circ c)_* \mathbb{C}_{\mathcal{C}}$ is $g^* \mathbb{V}_\chi$. Thus, by the isomorphism $g^* R^1 (e \circ c)_* \mathbb{C}_{\mathcal{C}} \cong \mathbb{H}$ on U' and Construction-Proposition 4.1(4),

$$\mathbb{H}_\chi = g^* \mathbb{V}_\chi \quad \text{and} \quad F^1 \mathbb{H}_\chi = g^* \mathcal{A}.$$

Thus $(F^1 \mathbb{H}_\chi)_Z = \mathcal{A}_Z$, the Schmid extension of $g^* \mathcal{A}$, and by Proposition 3.3, $\mathbf{M}$ descends to Z and

$$\mathcal{O}_Z(m M_Z) \cong \mathcal{A}_Z^{\otimes m}$$

for some positive integer m. By Set-up 3.1(5), $\mathbb{H}$ has unipotent local monodromy around $Z \backslash U'$, and $g^* \mathbb{V} \otimes \mathbb{C} \subset \mathbb{H}$, so $g^* \mathbb{V}$ has unipotent local monodromy around $Z \backslash U'$. By Proposition 5.1 for the composite $U' \to U_0 \subset U$, $\mathcal{A}_Z$ is nef and not semi-ample. Thus M_Z is nef and not semi-ample, which is (3).

Since $\mathbf{M}$ descends to Z and M_Z is not semi-ample, $\mathbf{M}_{Z'} = f^* M_Z$ is not semi-ample for any projective birational morphism $f : Z' \to Z$ from a normal variety, so $\mathbf{M}$ is not $\boldsymbol{b}$-semi-ample. □

7. Realizing the Hodge line bundle by sub-pairs on the weighted projective plane

The generic fiber of the fibration in Theorem 1.3 is a curve of genus 1, while Corollary 1.4 requires the generic fiber to be of Fano type (Definition 8.2).

Construction-Proposition 7.1. Notation as in Construction-Proposition 4.1. Let $S := \mathbb{P}(1, 2, 3)$ with weighted coordinates x, y, z of weights $1, 2, 3$, and $H := (x = 0)$. Then there exist a dominant generically finite morphism $g : U' \to U$ from a smooth variety U', a smooth projective variety Z containing U' as an open subset such that $Z \backslash U'$ is an snc divisor and $g^* \mathbb{V}$ has unipotent local monodromy around $Z \backslash U'$, and an lc-trivial fibration $\pi : (S \times Z, B) \to Z$ with B a $\mathbb{Q}$-divisor and π the second projection, satisfying the following.

(1) On the generic fiber of π, $B = E_S + \frac{1}{5}D_S - H$, where $E_S \in |6H|$ is a smooth curve and $D_S \in |5H|$ does not contain H.

(2) The moduli $\boldsymbol{b}$-divisor $\mathbf{M}$ of π descends to Z, and $\mathcal{O}_Z(mM_Z) \cong \mathcal{A}_Z^{\otimes m}$ for some positive integer m such that mM_Z is integral, where $\mathcal{A}_Z$ is the Schmid extension of $g^*\mathcal{A}$.

Proof. We divide the proof into five steps. Let $v_2 := [0:1:0]$, $v_3 := [0:0:1]$, $y_1 := y/x^2$, and $z_1 := z/x^3$. Thus $S\backslash H$ is the affine plane with coordinates y_1, z_1.

Step 1. The fibers of e are curves of genus 1 and P is a section of e, so the generic fiber of e has a Weierstrass equation with origin given by P. Thus there exist a dense open subset $U_1 \subset U$, regular functions a_4, a_6 on U_1, and an isomorphism ι over U_1 from $e^{-1}(U_1)$ onto

$$E_S := (\sigma_E = 0) \subset S \times U_1, \quad \text{where} \quad \sigma_E := z^2 - y^3 - a_4x^4y - a_6x^6,$$

such that $\iota(P \cap e^{-1}(U_1)) = \{[0:1:1]\} \times U_1$ and, for any $u \in U_1$, the restrictions of y_1 and z_1 to $\iota(E_u)$ have poles of orders 2 and 3 at $\iota(P_u)$. Setting $x = 0$ in σ_E, we obtain $E_S \cap (H \times U_1) = \{[0:1:1]\} \times U_1$. In particular, E_S is disjoint from $\{v_2, v_3\} \times U_1$. Near $[0:1:1]$, the functions x and z are local coordinates of S on the chart $y = 1$, and $y_1 = x^{-2}$ there. Since $y_1|_{\iota(E_u)}$ has a pole of order 2 at $\iota(P_u)$,

$$H|_{\iota(E_u)} = \iota(P_u) \quad \text{for any } u \in U_1.$$

The monomials of degree 5 in x, y, z are $x^5, x^3y, xy^2, x^2z, yz$. Their quotients by x^5 are $1, y_1, y_1^2, z_1, y_1z_1$, whose restrictions to $\iota(E_u)$ are regular outside $\iota(P_u)$ and have poles of orders $0, 2, 4, 3, 5$ at $\iota(P_u)$. Since $h^0(E_u, \mathcal{O}_{E_u}(5P_u)) = 5$, the five restrictions form a basis of $H^0(E_u, \mathcal{O}_{E_u}(5P_u))$ for any $u \in U_1$, so their pullbacks by ι form a basis of the locally free sheaf $e_*\mathcal{O}_{\mathcal{E}}(5P)|_{U_1}$. By Construction-Proposition 4.1(5), $\operatorname{div}(\xi) = D - 5P$, so ξ is a section of $\mathcal{O}_{\mathcal{E}}(5P)$ over $e^{-1}(U_1)$, and there exist regular functions $\lambda_1, \dots, \lambda_5$ on U_1 such that

$$\sigma_D := \lambda_1x^5 + \lambda_2x^3y + \lambda_3xy^2 + \lambda_4x^2z + \lambda_5yz \quad \text{satisfies} \quad \frac{\sigma_D}{x^5} \circ \iota = \xi \text{ on } e^{-1}(U_1).$$

We let $D_S := (\sigma_D = 0) \subset S \times U_1$. Since $\operatorname{div}(\xi) = D - 5P$ and $H|_{\iota(E_u)} = \iota(P_u)$, we have $D_S|_{E_S} = \iota(D \cap e^{-1}(U_1))$. Since $D_u \cap P_u = \emptyset$, $\xi|_{E_u}$ has a pole of order 5 at P_u. Among $1, y_1, y_1^2, z_1, y_1z_1$, only y_1z_1 has a pole of order 5 at $\iota(P_u)$, so λ_5 has no zero on U_1. Hence $\sigma_D|_{x=0} = \lambda_5yz$ and $D_S \cap (H \times U_1) = \{v_2, v_3\} \times U_1$. Moreover, on $(S\backslash H) \times U_1$,

$$\frac{\sigma_D}{x^5} = \lambda_5(y_1 - t_0)z_1 + \lambda_1 + \lambda_2y_1 + \lambda_3y_1^2, \quad \text{where} \quad t_0 := -\frac{\lambda_4}{\lambda_5}. \tag{7.1}$$

Step 2. Let $\mathbf{k}$ be an algebraic closure of $\mathbb{C}(U_1)$. In this step we work over $\mathbf{k}$, keep the notation S, H, E_S, D_S for the base changes to $\mathbf{k}$ of S, H and of the generic fibers of E_S, D_S over U_1, and let

$$B_S := E_S + \frac{1}{5}D_S - H.$$

By **Step 1**, E_S is a smooth curve contained in $S\backslash\{v_2, v_3\}$, $H|_{E_S} = [0:1:1]$, $D_S \cap H = \{v_2, v_3\}$, and $D_S|_{E_S}$ is reduced of degree 5, since D is étale over U. The open subset $(x \neq 0)$ of S is $\mathbb{A}^2$, and the open subsets $(y \neq 0)$ and $(z \neq 0)$ are the quotients of the affine plane with coordinates x, z by μ_2 acting by -1 and of the affine plane with coordinates x, y by μ_3 acting with weights $(1, 2)$. Hence $\operatorname{Sing} S = \{v_2, v_3\}$, and the singularities $v_2 \in S$ and $v_3 \in S$ are of types $\frac{1}{2}(1, 1)$ and $\frac{1}{3}(1, 2)$. By [Dol82, Theorem 3.3.4], $K_S \sim -6H$. Since $E_S \sim 6H$ and $D_S \sim 5H$,

$$K_S + B_S \sim_{\mathbb{Q}} (-6 + 6 + 1 - 1)H = 0.$$

We show that $\operatorname{Supp} B_S \cap S_{\mathrm{reg}}$ is an snc divisor, where $S_{\mathrm{reg}} := S \setminus \operatorname{Sing} S$. Since $S_{\mathrm{reg}} \cap H = H\backslash\{v_2, v_3\}$ is disjoint from D_S, we only need to consider $D_S \cap (S\backslash H)$. By (7.1), a singular point of $D_S \cap (S\backslash H)$ satisfies $y_1 = t_0$ and $\lambda_1 + \lambda_2t_0 + \lambda_3t_0^2 = 0$. Thus either $D_S \cap S_{\mathrm{reg}}$ is smooth, or $\lambda_1 + \lambda_2t_0 + \lambda_3t_0^2 = 0$ and

$$\frac{\sigma_D}{x^5} = (y_1 - t_0)(\lambda_5z_1 + \lambda_3y_1 + \lambda_2 + \lambda_3t_0),$$

in which case $D_S \cap S_{\mathrm{reg}}$ is the union of two smooth curves intersecting transversally at one point of $S \backslash H$. Since $D_S|_{E_S}$ is reduced, E_S intersects D_S transversally and $E_S \cap \operatorname{Sing} D_S = \emptyset$. Since $H|_{E_S} = [0:1:1]$ and $[0:1:1] \notin D_S$, E_S intersects H transversally at a point outside D_S. Hence $\operatorname{Supp} B_S \cap S_{\mathrm{reg}}$ is an snc divisor.

The singularities v_2, v_3 are Du Val. Let $q : \widetilde{S} \to S$ be the minimal resolution, Θ_1 the exceptional curve over v_2, and Θ_2, Θ_3 the exceptional curves over v_3. Then $\Theta_1^2 = \Theta_2^2 = \Theta_3^2 = -2$ and $\Theta_2 \cdot \Theta_3 = 1$. Near v_2 and v_3, q is the toric resolution in the coordinates x, z and x, y, and for $\Gamma_1 := (z = 0)$ and $\Gamma_2 = \Gamma_3 := (y = 0)$,

$$(\operatorname{mult}_{\Theta_1} q^*H, \operatorname{mult}_{\Theta_1} q^*\Gamma_1) = (\tfrac{1}{2}, \tfrac{1}{2}),$$
$$(\operatorname{mult}_{\Theta_2} q^*H, \operatorname{mult}_{\Theta_2} q^*\Gamma_2) = (\tfrac{1}{3}, \tfrac{2}{3}),$$
$$\text{and } (\operatorname{mult}_{\Theta_3} q^*H, \operatorname{mult}_{\Theta_3} q^*\Gamma_3) = (\tfrac{2}{3}, \tfrac{1}{3}).$$

Hence, for a curve $\Lambda \subset S$ and $i \in \{1, 2, 3\}$,

$$\operatorname{mult}_{\Theta_i} q^*\Lambda = \min\{a \operatorname{mult}_{\Theta_i} q^*H + b \operatorname{mult}_{\Theta_i} q^*\Gamma_i\},$$

where the minimum runs over the monomials $x^a z^b$ (for $i = 1$) or $x^a y^b$ (for $i \in \{2, 3\}$) of a local equation of the preimage of Λ in the affine plane near v_2 (for $i = 1$) or v_3 (for $i \in \{2, 3\}$). The preimages of H and D_S are defined by x and by

$$\lambda_3 x + \lambda_5 z + \lambda_4 x^2 z + \lambda_2 x^3 + \lambda_1 x^5 \text{ near } v_2, \qquad \lambda_5 y + \lambda_4 x^2 + \lambda_3 x y^2 + \lambda_2 x^3 y + \lambda_1 x^5 \text{ near } v_3.$$

Since $\lambda_5 \neq 0$, we obtain $(\operatorname{mult}_{\Theta_i} q^*D_S)_{i=1}^3 = (\frac{1}{2}, \frac{2}{3}, \frac{1}{3})$ and $(\operatorname{mult}_{\Theta_i} q^*H)_{i=1}^3 = (\frac{1}{2}, \frac{1}{3}, \frac{2}{3})$. Let $D' := q_*^{-1} D_S$ and $H' := q_*^{-1} H$, and denote $q^{-1}(E_S) \cong E_S$ again by E_S. Since q is crepant, $K_{\widetilde{S}} + \widetilde{B} = q^*(K_S + B_S)$ for

$$\widetilde{B} := q^*B_S = E_S + \frac{1}{5}D' - H' - \frac{2}{5}\Theta_1 - \frac{1}{5}\Theta_2 - \frac{3}{5}\Theta_3.$$

From $q^*D_S \cdot \Theta_i = q^*H \cdot \Theta_i = 0$ for any $i \in \{1, 2, 3\}$ we obtain

$$D' \cdot \Theta_1 = D' \cdot \Theta_2 = H' \cdot \Theta_1 = H' \cdot \Theta_3 = 1 \quad \text{and} \quad D' \cdot \Theta_3 = H' \cdot \Theta_2 = 0.$$

Near v_2, the preimages of D_S and H are smooth with the distinct tangent lines $(\lambda_3 x + \lambda_5 z = 0)$ and $(x = 0)$, and Θ_1 is the curve of tangent lines at the origin, so $D' \cap \Theta_1 \neq H' \cap \Theta_1$. Near v_3, $D' \cap \Theta_3 = H' \cap \Theta_2 = \emptyset$ since $D' \cdot \Theta_3 = H' \cdot \Theta_2 = 0$, and D' intersects Θ_2 and H' intersects Θ_3 transversally at one point since $D' \cdot \Theta_2 = H' \cdot \Theta_3 = 1$. Since moreover $D_S \cap H = \{v_2, v_3\}$ and $\operatorname{Supp} B_S \cap S_{\mathrm{reg}}$ is snc, $\operatorname{Supp} \widetilde{B}$ is an snc divisor.

Since $(\widetilde{S}, \widetilde{B})$ is log smooth with coefficients at most 1, it is sub-lc, so (S, B_S) is sub-lc. Hence $\lceil \mathbf{A}^*(S, B_S) \rceil \geq 0$ by Definition 2.3, so the constant function 1 is a section of $\mathcal{O}(\lceil \mathbf{A}^*(S, B_S) \rceil)$ on any higher model of S. Conversely, the trace of $\lceil \mathbf{A}^*(S, B_S) \rceil$ on S is H, because $a(E_S, S, B_S) = 0$, $a(H, S, B_S) = 2$, $a(\Lambda, S, B_S) = \frac{4}{5}$ for any component Λ of D_S, and $a(\Lambda, S, B_S) = 1$ for any other prime divisor Λ on S. So any section is a rational function f on S with $\operatorname{div}(f) + H \geq 0$, that is, $f \in x^{-1} H^0(S, \mathcal{O}_S(1)) = \mathbf{k}$. Hence

$$\dim_{\mathbf{k}} H^0(S, \mathcal{O}_S(\lceil \mathbf{A}^*(S, B_S) \rceil)) = 1.$$

Step 3. Let $B_{S \times U_1} := E_S + \frac{1}{5} D_S - H \times U_1$ and

$$\phi := \frac{x^{35}}{\sigma_E^5 \sigma_D}.$$

Since $\operatorname{div}(\sigma_E / x^6) = E_S - 6H \times U_1$ and $\operatorname{div}(\sigma_D / x^5) = D_S - 5H \times U_1$, we have

$$\operatorname{div}(\phi) = 35H \times U_1 - 5E_S - D_S,$$

that is,

$$5(-6H \times U_1 + B_{S \times U_1}) + \operatorname{div}(\phi) = 0. \tag{7.2}$$

Let $w := (x^6/\sigma_E)\phi^{-1/5}$, a rational function on the cyclic cover of $S \times U_1$ given by a fifth root of ϕ. Then $w^5 = \sigma_D/x^5$ and $\phi^{1/5} = x^6/(\sigma_E w)$. By (7.1),

$$z_1 = \frac{w^5 - \lambda_1 - \lambda_2 y_1 - \lambda_3 y_1^2}{\lambda_5(y_1 - t_0)},$$

so, over $y_1 \neq t_0$, the hypersurface

$$(w^5 = \sigma_D/x^5) \subset \mathbb{A}^1 \times (S\backslash H) \times U_1$$

is the graph of z_1 as a function of (y_1, w) and the point of U_1. Since

$$\dim\big((w^5 = \sigma_D/x^5) \cap (y_1 = t_0)\big) = \dim U_1 + 1,$$

no component of $(w^5 = \sigma_D/x^5)$ is contained in $(y_1 = t_0)$. Thus the hypersurface $(w^5 = \sigma_D/x^5)$ is irreducible and birational to $\mathbb{A}^2 \times U_1$ via (y_1, w) and the projection to U_1. Since $\phi^{1/5} = x^6/(\sigma_E w)$, the cyclic cover of $S \times U_1$ given by a fifth root of ϕ is birational to $(w^5 = \sigma_D/x^5)$ via w.

Step 4. In this step we construct g, Z, and B, prove (1), and show that $\mathbf{M}$ descends to Z with $\mathcal{O}_Z(mM_Z) \cong ((F^2\mathbb{H}_\chi)_Z)^{\otimes m}$.

Let $B_{\widetilde{S}\times U_1} := (q \times \mathrm{id}_{U_1})^* B_{S\times U_1}$. The base change of $B_{\widetilde{S}\times U_1}$ to the geometric generic fiber is $\widetilde{B}$. By **Step 2** and generic smoothness, there exists a dense open subset $U_2 \subset U_1$ such that, for

$$B_{\widetilde{S}\times U_2} := B_{\widetilde{S}\times U_1}|_{\widetilde{S}\times U_2},$$

$\operatorname{Supp} B_{\widetilde{S}\times U_2}$ is an snc divisor. Possibly shrinking U_2, we may assume that $\Omega^5_{U_2} \cong \mathcal{O}_{U_2}$. We fix a generator ω of $\Omega^5_{U_2}$, so that $K_{U_2} := 0$ is a canonical divisor of U_2. Let $K_{\widetilde{S}} := q^*(-6H)$, a canonical divisor of $\widetilde{S}$ since q is crepant, and $K_{\widetilde{S}\times U_2} := K_{\widetilde{S}} \times U_2$. By (7.2),

$$5(K_{\widetilde{S}\times U_2} + B_{\widetilde{S}\times U_2}) + \operatorname{div}(\phi) = 0,$$

where ϕ also denotes the pullback of ϕ. Let X_η be the generic fiber of $\widetilde{S} \times U_2 \to U_2$. Since D' has coefficient $\frac{1}{5}$, $k(K_{X_\eta} + B_{\widetilde{S}\times U_2}|_{X_\eta})$ is integral only if $5 \mid k$, so by (7.2),

$$\min\{k \in \mathbb{N}^+ \mid k(K_{X_\eta} + B_{\widetilde{S}\times U_2}|_{X_\eta}) \sim 0\} = 5.$$

By Lemma 3.4 for $\widetilde{S} \times U_2 \to U_2$, $B_{\widetilde{S}\times U_2}$, and ϕ, there exist a dense open subset $U_3 \subset U_2$, a finite étale morphism $g : U' \to U_3$ from a smooth variety U', and a smooth projective variety Z containing U' as an open subset, with the property in Lemma 3.4. We also regard g as a morphism to U. Then g is dominant and generically finite.

Since g is étale, $g^*\omega$ is a generator of $\Omega^5_{U'}$. We let K_Z be the divisor of $g^*\omega$ on Z, so that $K_Z|_{U'} = 0$, and let $K_{S\times Z} := -6H \times Z + S \times K_Z$. Let B^h be the closure in $S \times Z$ of the pullback of $B_{S\times U_2} := B_{S\times U_1}|_{S\times U_2}$ under $\mathrm{id}_S \times g$. Since S has only quotient singularities, $S \times Z$ is $\mathbb{Q}$-factorial. By (7.2) and $K_Z|_{U'} = 0$, the divisor $5(K_{S\times Z} + B^h) + \operatorname{div}(\phi)$ is zero on $S \times U'$, so no component of $5(K_{S\times Z} + B^h) + \operatorname{div}(\phi)$ dominates Z. Since U' is dense in Z and $(S \times U', B^h|_{S\times U'})$ is the base change of $(S \times U_2, B_{S\times U_2})$ by the finite morphism g (we omit the passage from $\mathbf{k}$, an algebraic closure of $\mathbb{C}(Z)$, to $\mathbb{C}(Z)$ by flat base change on the log resolution $\widetilde{S}$ of **Step 2**), **Step 2** shows that $(S \times Z, B^h)$ is sub-lc over the generic point of Z and

$$\operatorname{rank} \pi_* \mathcal{O}_{S\times Z}(\lceil \mathbf{A}^*(S \times Z, B^h)\rceil) = 1.$$

By Lemma 3.5, $\pi : (S \times Z, B) \to Z$ is an lc-trivial fibration with $K_{S\times Z} + B \sim_{\mathbb{Q}} 0$, where

$$B := B^h + B^v \quad \text{and} \quad B^v := -\frac{1}{5}\big(5(K_{S\times Z} + B^h) + \operatorname{div}(\phi)\big),$$

and $B^v|_{S\times U'} = 0$. On the generic fiber of π, $B = B^h = E_S + \frac{1}{5}D_S - H$, where $E_S \in |6H|$ is smooth and $D_S \in |5H|$ does not contain H by **Step 1**. Thus (1) holds.

Let $\widetilde{\pi} : \widetilde{S} \times Z \to Z$ be the second projection, $B_{\widetilde{S}\times Z} := (q \times \mathrm{id}_Z)^* B$, and

$$K_{\widetilde{S}\times Z} := K_{\widetilde{S}} \times Z + \widetilde{S} \times K_Z.$$

Since $q \times \mathrm{id}_Z$ is crepant, $\widetilde{\pi}$ is an lc-trivial fibration for the sub-pair $(\widetilde{S} \times Z, B_{\widetilde{S}\times Z})$, and the discriminant and moduli $\boldsymbol{b}$-divisors of $\widetilde{\pi}$ are those of π. Since $B^v|_{S\times U'} = 0$, the sub-pair

$$(\widetilde{\pi}^{-1}(U'), B_{\widetilde{S}\times Z}|_{\widetilde{\pi}^{-1}(U')})$$

is the base change of $(\widetilde{S} \times U_2, B_{\widetilde{S}\times U_2})$ by g. By Lemma 3.4, the conditions of Set-up 3.1 hold for $\widetilde{\pi}$, $\Sigma := Z\backslash U'$, the pullbacks of ϕ and $K_{\widetilde{S}\times U_2}$, and some μ_5-equivariant resolution h, with $d = 2$ and $r = 5$. In particular $Z\backslash U'$ is an snc divisor. By Proposition 3.3, $\mathbf{M}$ descends to Z and

$$\mathcal{O}_Z(mM_Z) \cong ((F^2\mathbb{H}_\chi)_Z)^{\otimes m}$$

for some positive integer m such that mM_Z is integral, where $\mathbb{H}_\chi$ is the local system of Set-up 3.1 for $\widetilde{\pi}$.

Step 5. In this step we show that $(F^2\mathbb{H}_\chi)_Z \cong \mathcal{A}_Z$ and that $g^*\mathbb{V}$ has unipotent local monodromy around $Z\backslash U'$.

We use the notation of Set-up 3.1 for $\widetilde{\pi}$: $Z^\circ = U'$, $X^\circ = \widetilde{S} \times U'$, $f : X' \to X^\circ$, $\pi'' : X'' \to U'$, and E''. We keep the notation E_S, D', H', ι, w for their base changes by g. Then $(B^\circ)^{=1} = E_S$. Near E_S, we have $5(K_{X^\circ} + B^\circ) = -35H' \times U' + 5E_S + D'$ and $D'|_{E_S} = \iota(D)$, and the divisor D' is smooth and intersects E_S transversally since $\iota(D)$ is étale over U'. Hence X' is smooth near $f^{-1}(E_S)$, f is flat there, and $f^{-1}(E_S) \to E_S$ is a cyclic cover of degree 5 branched along $\iota(D)$. Since $w^5 = \sigma_D/x^5$ and $(\sigma_D/x^5) \circ \iota = \xi$, Construction-Proposition 4.1(5) implies that ι lifts to an isomorphism ι' over U' from $\mathcal{C} \times_U U'$ onto $f^{-1}(E_S)$. We let μ_5 act on X' so that ι' is μ_5-equivariant. The new action differs from the action of Set-up 3.1(4) by an automorphism of μ_5, so h is still μ_5-equivariant and $\mathbb{H}_\chi$ does not change. Under ι', w corresponds to $\xi^{1/5}$. Since $\phi^{1/5} = x^6/(\sigma_E w)$ and x^6/σ_E is a rational function on X°, Construction-Proposition 4.1(5) shows that μ_5 acts on $\phi^{1/5}$ by χ, so the character χ of Set-up 3.1 is the character χ of Construction-Proposition 4.1. For $u \in U'$, ι' restricts to an isomorphism from $C_{g(u)}$ onto the fiber of $f^{-1}(E_S)$ over u.

Let $u \in U'$ be a general point. By **Step 3**, the rational map $X'' \dashrightarrow \mathbb{A}^2 \times U'$ given by (y_1, w) and π'' is birational, so X''_u is a smooth projective rational surface. Thus $H^1(X''_u) = H^3(X''_u) = 0$ and $H^2(X''_u)$ is of type $(1, 1)$. Since X'_u is smooth near $C_{g(u)}$, $h_u : X''_u \to X'_u$ is near $C_{g(u)}$ a composition of blow-ups of points. Hence $E''_u = h_u^{-1}(C_{g(u)})$ consists of $(h_u)_*^{-1}C_{g(u)} \cong C_{g(u)}$ and of smooth rational curves, and the dual graph of E''_u is a tree. Let $E''^{[k]}_u$ be the disjoint union of the k-fold intersections of components of E''_u. By the weight spectral sequence [Del71, Section 3.2], the weights of $H^2(X''_u\backslash E''_u)$ are in $\{2, 3, 4\}$ and

$$\begin{aligned} &\mathrm{gr}_2^W H^2(X''_u\backslash E''_u) = \mathrm{Im}\big(H^2(X''_u) \to H^2(X''_u\backslash E''_u)\big),\\ &\mathrm{gr}_3^W H^2(X''_u\backslash E''_u) \cong H^1(C_{g(u)})(-1),\\ \text{and}\quad &\mathrm{gr}_4^W H^2(X''_u\backslash E''_u) = 0. \end{aligned}$$

Here the isomorphism is the residue along $(h_u)_*^{-1}C_{g(u)}$, which is an isomorphism because H^1 of a rational curve and $H^3(X''_u)$ vanish, and $\mathrm{gr}_4^W = 0$ because the Gysin map $H^0(E''^{[2]}_u) \to H^2(E''^{[1]}_u)$ is injective for a tree. The residue and the Gysin maps are μ_5-equivariant. By Set-up 3.1(4), the strata of E'' are smooth over U', so W is a filtration of $\mathbb{H}$ by sub-local systems, the residue is a morphism of variations of mixed Hodge structure, and the three descriptions of gr_k^W above hold for the variations over U'. Since $H^1(C_{g(u)}) = H^1(E_{g(u)}) \oplus \mathbb{V}_{g(u)}$ with μ_5 acting trivially on $H^1(E_{g(u)})$, and χ is non-trivial, we obtain

$$\mathbb{H}_\chi = W_3\mathbb{H}_\chi, \qquad F^2\,\mathrm{gr}_2^W\,\mathbb{H}_\chi = 0, \qquad \text{and} \qquad \mathrm{gr}_3^W\,\mathbb{H}_\chi \cong g^*\mathbb{V}_\chi(-1).$$

Since $\mathbb{H}_\chi = W_3\mathbb{H}_\chi$, the filtration F on $\mathrm{gr}_3^W\,\mathbb{H}_\chi$ is the image of F on $\mathbb{H}_\chi$. Since moreover

$$F^2W_2\mathbb{H}_\chi = F^2\,\mathrm{gr}_2^W\,\mathbb{H}_\chi = 0,$$

the projection to $\mathrm{gr}_3^W \mathbb{H}_\chi$ restricts to an isomorphism

$$F^2\mathbb{H}_\chi \to F^2 \mathrm{gr}_3^W \mathbb{H}_\chi.$$

By the equality $(F^2\mathbb{H}_\chi)_Z = F^2(\mathcal{V}_Y^{\mathrm{tr}})_\chi$ in [BFMT25, Construction-Definition 6.23], where $\mathcal{V}_Y^{\mathrm{tr}}$ is the Deligne extension [BFMT25, Section 2.4] of the transcendental part V_Y^{tr} and the χ-isotypic part of V_Y^{tr} is a sub-variation of $\mathrm{gr}_3^W \mathbb{H}_\chi$ containing $F^2 \mathrm{gr}_3^W \mathbb{H}_\chi$ [BFMT25, Definition 2.9(3) and Remark 6.26] whose Deligne extension is a subbundle of that of $\mathrm{gr}_3^W \mathbb{H}_\chi$ [Del70, Chapter II, Proposition 5.2(d)], the Schmid extension $(F^2\mathbb{H}_\chi)_Z$ is the Schmid extension of $F^2 \mathrm{gr}_3^W \mathbb{H}_\chi$ in $\mathrm{gr}_3^W \mathbb{H}_\chi$. A Tate twist does not change the underlying line bundle, and

$$F^2 \mathrm{gr}_3^W \mathbb{H}_\chi \cong F^2(g^*\mathbb{V}_\chi(-1)) = g^*\mathbb{V}_\chi^{1,0} = g^*\mathcal{A}.$$

Thus

$$(F^2\mathbb{H}_\chi)_Z \cong \mathcal{A}_Z,$$

and by **Step 4**, $\mathcal{O}_Z(mM_Z) \cong \mathcal{A}_Z^{\otimes m}$, which is (2).

By Set-up 3.1(5), $R^2\pi''_*\mathbb{Z}_{X''\setminus E''}$ has unipotent local monodromy around $Z\setminus U'$. Hence the subquotient gr_3^W of $R^2\pi''_*\mathbb{Q}_{X''\setminus E''}$ has unipotent local monodromy around $Z\setminus U'$. The residue is an isomorphism

$$\mathrm{gr}_3^W R^2\pi''_*\mathbb{Q}_{X''\setminus E''} \to g^*R^1(e\circ c)_*\mathbb{Q}_{\mathcal{C}}(-1),$$

and $g^*\mathbb{V}(-1)$ is a sub-local system of the right-hand side, so $g^*\mathbb{V}(-1)$ has unipotent local monodromy around $Z\setminus U'$. Thus $g^*\mathbb{V}$ has unipotent local monodromy around $Z\setminus U'$. □

8. Proof of Corollary 1.4

Definition 8.1 (cf. [HJL22, Definitions 2.5 and 2.7])**.** Let $(X/Z \ni z, B)$ be a sub-pair with contraction $\pi : X \to Z$, and let $\overline{z}$ be the closure of z. The *minimal log discrepancy* of $(X/Z \ni z, B)$ is

$$\mathrm{mld}(X/Z \ni z, B) := \inf\{a(E, X, B) \mid E \text{ is a prime divisor over } X \text{ with } \pi(\mathrm{center}_X E) = \overline{z}\}.$$

For a real number $\delta \geq 0$, the sub-pair $(X/Z \ni z, B)$ is *δ-lc* if $\mathrm{mld}(X/Z \ni z, B) \geq \delta$, and a sub-pair (X, B) is *δ-lc* if $(X \ni x, B)$ is δ-lc for any (not necessarily closed) point $x \in X$ of codimension ≥ 1.

Definition 8.2 (cf. [HJL22, Definition 2.16])**.** A normal projective variety X over a field $k \supset \mathbb{C}$ is *of Fano type* if there exists a klt pair (X, B) over k (Definition 2.1 with $\mathbb{C}$ replaced by k) such that $-(K_X + B)$ is ample.

Proof of Corollary 1.4. We divide the proof into several steps.

Step 1. Let $g : U' \to U$ and $\pi_W : (S \times W, \Delta) \to W$ be the morphism and the lc-trivial fibration given by Construction-Proposition 7.1, where W and Δ are the smooth projective variety and the $\mathbb{Q}$-divisor denoted by Z and B there. Let B_W and M_W be the discriminant and the moduli part of π_W, and let L_W be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor as in Definition 2.5. Since $\dim U = 5$ and g is dominant and generically finite, $\dim W = 5$. By Construction-Proposition 7.1(2) and Proposition 5.1, $\mathbf{M}$ descends to W, and M_W is nef, not big, and not semi-ample.

Since M_W is not semi-ample, $\mathrm{Bs}|knM_W| \subset \mathrm{Bs}|nM_W|$ for any $k \in \mathbb{N}^+$, and W is quasi-compact, there exists a closed point $p \in W$ satisfying

$$p \in \mathrm{Bs}|nM_W| \quad \text{for any positive integer } n \text{ such that } nM_W \text{ is integral.} \tag{8.1}$$

Next we show that for any $\mathbb{R}$-divisor $D \geq 0$ with $D \sim_{\mathbb{R}} M_W$ we have $p \in \mathrm{Supp}\, D$. Write

$$D = \sum_{i=1}^{N} \gamma_i D_i \quad \text{and} \quad D - M_W = \sum_{j=1}^{N'} \lambda_j \,\mathrm{div}(\psi_j),$$

where $D_1, \dots, D_N$ are distinct prime divisors, $\gamma_i > 0$, $\lambda_j \in \mathbb{R}$, and ψ_j are rational functions on W. Comparing the coefficients along the finitely many prime divisors contained in

$$\operatorname{Supp} D \cup \operatorname{Supp} M_W \cup \bigcup_{j=1}^{N'} \operatorname{Supp} \operatorname{div}(\psi_j),$$

the vector $(\gamma_1, \dots, \gamma_N, \lambda_1, \dots, \lambda_{N'})$ is a real solution of a system of linear equations with rational coefficients. Since the rational solutions of such a system are dense in the real solutions, there exists a rational solution $(\gamma'_1, \dots, \gamma'_N, \lambda'_1, \dots, \lambda'_{N'})$ with $\gamma'_i > 0$ for any i. Let n_0 be a positive integer such that $n_0 M_W$, $n_0\gamma'_i$, and $n_0\lambda'_j$ are integral for any i and j. Then

$$\sum_{i=1}^{N} n_0\gamma'_i D_i = n_0 M_W + \operatorname{div}\Big(\prod_{j=1}^{N'} \psi_j^{n_0\lambda'_j}\Big) \in |n_0 M_W|,$$

so $p \in \bigcup_{i=1}^{N} D_i = \operatorname{Supp} D$ by (8.1).

Step 2. Let A be a very ample Cartier divisor on W. Since M_W is nef, there exists a positive integer m such that $K_W + 5mA + M_W$ is ample. By Bertini, there exist $R_1, \dots, R_5 \in |mA|$ containing p such that local equations of $R_1, \dots, R_5$ at p form a regular system of parameters of W at p and $R_1 + \cdots + R_5$ is snc. Since $R_1 + \cdots + R_5 \sim 5mA$, $K_W + R_1 + \cdots + R_5 + M_W$ is ample. Let r be a positive integer such that $r(K_W + R_1 + \cdots + R_5 + M_W)$ is integral and very ample, let F' be a general member of $|r(K_W + R_1 + \cdots + R_5 + M_W)|$, and let $F := \frac{1}{r}F'$. Then $p \notin \operatorname{Supp} F$, $R_1 + \cdots + R_5 + F$ is snc, and we let

$$B'_W := R_1 + \cdots + R_5 - F.$$

Since $F' \in |r(K_W + R_1 + \cdots + R_5 + M_W)|$, there exists a rational function φ on W with

$$r(K_W + B'_W + M_W) = \operatorname{div}_W(\varphi). \tag{8.2}$$

Since B'_W is snc with coefficients in $\{1, -\frac{1}{r}\}$, (W, B'_W) is sub-lc.

Let

$$\Delta' := \Delta + \pi_W^*(B'_W - B_W) \quad \text{and} \quad L'_W := L_W + B'_W - B_W = K_W + B'_W + M_W.$$

Then Δ' is a $\mathbb{Q}$-divisor and $K_{S\times W} + \Delta' \sim_{\mathbb{Q}} \pi_W^* L'_W$. Since $\Delta' = \Delta$ over $W \setminus \operatorname{Supp}(B'_W - B_W)$, π_W with the boundary Δ' is an lc-trivial fibration, which we denote by $\pi'_W : (S \times W, \Delta') \to W$, and we take L'_W as L in Definition 2.5. Let $f : W' \to W$ be a projective birational morphism from a normal variety. Since the pullbacks of $K_{S\times W} + \Delta'$ and $K_{S\times W} + \Delta$ to any higher model differ by the pullback of $f^*(B'_W - B_W)$, for any prime divisor P on W' the threshold t_P of Definition 2.5 for Δ' is the threshold for Δ minus $\operatorname{mult}_P f^*(B'_W - B_W)$. Let $\mathbf{B}$ and $\mathbf{B}'$ be the discriminant $\boldsymbol{b}$-divisors of π_W and π'_W, and $\mathbf{M}'$ the moduli $\boldsymbol{b}$-divisor of π'_W. Then $\mathbf{B}'_{W'} = \mathbf{B}_{W'} + f^*(B'_W - B_W)$, and since $\mathbf{M}$ descends to W,

$$\mathbf{M}'_{W'} = f^*L'_W - K_{W'} - \mathbf{B}'_{W'} = f^*L_W - K_{W'} - \mathbf{B}_{W'} = f^*M_W.$$

Thus the discriminant and the moduli $\boldsymbol{b}$-divisor of π'_W are B'_W and $\mathbf{M}$, so the moduli part of π'_W is M_W. Moreover, for any projective birational morphism $f : W' \to W$ from a normal variety,

$$K_{W'} + \mathbf{B}'_{W'} = f^*(K_W + B'_W). \tag{8.3}$$

We show that $(S \times W, \Delta')$ is sub-lc. Let v be a prime divisor over $S \times W$. If the center of v on $S \times W$ dominates W, then $a(v, S \times W, \Delta') \geq 0$ since $(S \times W, \Delta')$ is sub-lc over the generic point of W.

Suppose that the center of v on $S \times W$ does not dominate W. Let $X_1 \to S \times W$ be a projective birational morphism from a normal variety X_1 such that v is a prime divisor on X_1. The induced morphism $X_1 \to W$ is projective and surjective, and the generic fiber of $X_1 \to W$ is geometrically integral since it is birational to the generic fiber of the second projection π_W and S is a variety over the algebraically closed field $\mathbb{C}$. By [AK00, Theorem 2.1 and Proposition 4.4], there exist

projective modifications $X_2 \to X_1$ and $W_2 \to W$ and an equidimensional toroidal morphism $\pi_2 : X_2 \to W_2$ compatible with $X_1 \to W$. Since X_1 is normal, $X_2 \to X_1$ is an isomorphism over the generic point of v, so v is a prime divisor on X_2. Since π_2 is equidimensional,

$$\dim \pi_2(v) \geq \dim v - (\dim X_2 - \dim W_2) = \dim W_2 - 1,$$

and since v does not dominate W_2, $P := \pi_2(v)$ is a prime divisor on W_2 and $\operatorname{mult}_v \pi_2^* P$ is a positive integer. Let $K_{X_2} + \Delta_2$ be the pullback of $K_{S\times W} + \Delta'$ to X_2, and let t_P be the supremum of the real numbers t such that $(X_2, \Delta_2 + t\pi_2^* P)$ is sub-lc over the generic point of P. Since X_2 is birational over W_2 to the main component of $(S \times W) \times_W W_2$, $1 - t_P$ is the coefficient of P in the trace $\mathbf{B}'_{W_2}$ of the discriminant $\boldsymbol{b}$-divisor of π'_W. By (8.3) for $W_2 \to W$ and since (W, B'_W) is sub-lc,

$$1 - t_P = \operatorname{mult}_P \mathbf{B}'_{W_2} = 1 - a(P, W, B'_W) \leq 1,$$

so $t_P \geq 0$. For any real number $t < t_P$, we have

$$a(v, X_2, \Delta_2) - t \operatorname{mult}_v \pi_2^* P = a(v, X_2, \Delta_2 + t\pi_2^* P) \geq 0,$$

so

$$a(v, S \times W, \Delta') = a(v, X_2, \Delta_2) \geq t_P \operatorname{mult}_v \pi_2^* P \geq 0.$$

Thus $(S \times W, \Delta')$ is sub-lc. Moreover, by (8.2),

$$K_{S\times W} + \Delta' \sim_{\mathbb{Q}} \pi_W^*(K_W + B'_W + M_W) \sim_{\mathbb{Q}} 0.$$

Step 3. Let

$$Y := \operatorname{Spec}_W \bigoplus_{i\geq 0} \mathcal{O}_W(iA) \quad \text{and} \quad Z_0 := \operatorname{Spec} \bigoplus_{i\geq 0} H^0(W, iA),$$

so that Y is the total space of $\mathcal{O}_W(-A)$, with projection $q : Y \to W$ and zero section $E \cong W$. By [Kol13, 3.8], Z_0 is normal and the natural projective birational morphism $\mu : Y \to Z_0$ contracts E to the vertex o of Z_0 and is an isomorphism outside E. The variety Y is smooth, $\mathcal{O}_Y(E) \cong q^*\mathcal{O}_W(-A)$, and the relative cotangent sheaf of q is $q^*\mathcal{O}_W(A)$, so we may take canonical divisors with

$$K_Y + E = q^* K_W. \tag{8.4}$$

Let $X_0 := S \times Y$, let $\pi_Y : X_0 \to Y$ be the second projection, let $\pi_0 := \mu \circ \pi_Y : X_0 \to Z_0$, and let

$$B_0 := (\mathrm{id}_S \times q)^* \Delta' + S \times E.$$

Since $X_0 \to S \times W$ is the total space of the pullback of $\mathcal{O}_W(-A)$ with zero section $S \times E$, and $K_{X_0} + S \times E = (\mathrm{id}_S \times q)^* K_{S\times W}$ as in (8.4), (X_0, B_0) is sub-lc by **Step 2**. By (8.4) and **Step 2**,

$$K_{X_0} + B_0 = (\mathrm{id}_S \times q)^*(K_{S\times W} + \Delta') \sim_{\mathbb{Q}} 0.$$

Since π_Y and μ are contractions, π_0 is a contraction, and $\dim X_0 - \dim Z_0 = 2$. For any open subset $W_1 \subset W$ on which $\mathcal{O}_W(-A)$ is trivial,

$$q^{-1}(W_1) \backslash E \cong W_1 \times (\mathbb{A}^1 \backslash \{0\}),$$

over which $\pi_Y : (X_0, B_0) \to Y$ is the product of the restriction of π'_W over W_1 with $\mathbb{A}^1\backslash\{0\}$. Since π'_W is an lc-trivial fibration and μ is an isomorphism outside E, the product structure implies $\operatorname{rank} \pi_{0*}\mathcal{O}_{X_0}(\lceil \mathbf{A}^*(X_0, B_0) \rceil) = 1$.

Let $B_Y := E + q^* B'_W$. By (8.2) and (8.4),

$$K_Y + B_Y + q^* M_W = q^*(K_W + B'_W + M_W) = \frac{1}{r} \operatorname{div}_Y(\varphi). \tag{8.5}$$

Let $K_{Z_0} := \mu_* K_Y$, $B_{Z_0} := \mu_* q^* B'_W$, and $M_{Z_0} := \mu_* q^* M_W$. Applying μ_* to (8.5), we obtain

$$L_0 := K_{Z_0} + B_{Z_0} + M_{Z_0} = \frac{1}{r} \operatorname{div}_{Z_0}(\varphi),$$

so L_0 is $\mathbb{Q}$-Cartier, $K_{X_0}+B_0 \sim_{\mathbb{Q}} 0 \sim_{\mathbb{Q}} \pi_0^* L_0$, and

$$K_Y + B_Y + q^* M_W = \mu^*(K_{Z_0} + B_{Z_0} + M_{Z_0}). \tag{8.6}$$

Thus $\pi_0 : (X_0, B_0) \to Z_0$ is an lc-trivial fibration with B_0 a $\mathbb{Q}$-divisor, and we take L_0 as L in Definition 2.5.

We show that B_Y is the trace on Y of the discriminant $\boldsymbol{b}$-divisor of π_0. Since π_0 factors through π_Y, the trace on Y of the discriminant $\boldsymbol{b}$-divisor of π_0 is the discriminant of $\pi_Y : (X_0, B_0) \to Y$. Let P be a prime divisor on Y and t_P the threshold of Definition 2.5 for π_Y.

- If $P = E$, then $\pi_Y^* E = S \times E$ has coefficient 1 in B_0 and (X_0, B_0) is sub-lc, so $t_E = 0$ and $\operatorname{mult}_E B_Y = 1$.
- If P does not dominate W, then $P = q^* P_W$ for the prime divisor $P_W := q(P)$ on W. The generic point of P is not in E, so by the product structure over $q^{-1}(W_1)\backslash E$ for such W_1 meeting P_W, $1 - t_P$ is the coefficient of P_W in the discriminant B'_W of π'_W, which is $\operatorname{mult}_P q^* B'_W$.
- If P dominates W and $P \neq E$, then over the generic point of W, π_Y is the product of the generic fiber of π'_W with the fiber of q, and P corresponds to a closed point of that fiber other than the zero point. Since the generic fiber of π'_W is sub-lc, $t_P = 1$ and $\operatorname{mult}_P B_Y = 0$.

Thus the trace on Y is B_Y. Since μ is an isomorphism outside E and E is μ-exceptional, the discriminant of π_0 is $\mu_* B_Y = B_{Z_0}$, and the moduli part of π_0 is

$$L_0 - K_{Z_0} - B_{Z_0} = M_{Z_0}.$$

Step 4. In this step we construct $\pi : (X, B) \to Z$ and z, and we prove (1)–(3).

Let s be the coordinate of $\mathbb{A}^1$, and let

$$Z := Z_0 \times \mathbb{A}^1, \quad z := (o, 0), \quad X := X_0 \times \mathbb{A}^1, \quad \text{and} \quad \pi := \pi_0 \times \mathrm{id}_{\mathbb{A}^1} : X \to Z.$$

Let B be the pullback of B_0 to X. By **Step 3**, $\pi : (X, B) \to Z$ is an lc-trivial fibration with B a $\mathbb{Q}$-divisor, z is a closed point of Z, and $\dim X - \dim Z = 2$. We take $L := \mathrm{pr}_{Z_0}^* L_0$ and $K_Z := \mathrm{pr}_{Z_0}^* K_{Z_0}$. For a prime divisor P on Z, either $P = \mathrm{pr}_{Z_0}^* P_0$ for a prime divisor P_0 on Z_0, and then $t_P = t_{P_0}$, or P dominates Z_0, and then $t_P = 1$ since over the generic point of Z_0, (X, B) is the product of the generic fiber of π_0 with $\mathbb{A}^1$ and $\pi^* P$ is reduced. Thus $B_Z = \mathrm{pr}_{Z_0}^* B_{Z_0}$ and $M_Z = \mathrm{pr}_{Z_0}^* M_{Z_0}$. Let $\mathrm{pr} : Y \times \mathbb{A}^1 \to Y$ be the first projection and $K_{Y\times\mathbb{A}^1} := \mathrm{pr}^* K_Y$. By (8.6),

$$(\mu \times \mathrm{id}_{\mathbb{A}^1})^*(K_Z + B_Z + M_Z) = K_{Y\times\mathbb{A}^1} + \mathrm{pr}^* B_Y + \mathrm{pr}^* q^* M_W. \tag{8.7}$$

By **Step 3**, (X_0, B_0) is sub-lc, so $(X, B + \pi^*(s = 0))$ is sub-lc. Let v be a prime divisor over X with $\pi(\operatorname{center}_X v) = \{z\}$. Since $\operatorname{center}_X v \subset \pi^{-1}(z) \subset \operatorname{Supp} \pi^*(s = 0)$, $\operatorname{mult}_v \pi^*(s = 0) \geq 1$, so

$$a(v, X, B) = a(v, X, B + \pi^*(s = 0)) + \operatorname{mult}_v \pi^*(s = 0) \geq 1.$$

Thus $\operatorname{mld}(X/Z \ni z, B) \geq 1$. The generic fiber of π is the base change of S to the generic point of Z, which is of Fano type (Definition 8.2) since S is klt and $-K_S$ is ample. Thus (1)–(3) hold.

Step 5. In this step we prove (4) and the last sentence of Corollary 1.4.

Let $V \subset Z$ be an open neighborhood of z and $D \geq 0$ an $\mathbb{R}$-divisor on V with $D \sim_{\mathbb{R}} M_Z|_V$, and write

$$D - M_Z|_V = \sum_{i \in I} \nu_i \operatorname{div}(\rho_i)$$

with I a finite set, $\nu_i \in \mathbb{R}$, and ρ_i rational functions on V. Let $V' := (\mu \times \mathrm{id}_{\mathbb{A}^1})^{-1}(V)$, which is smooth. Since $K_Z + B_Z + M_Z$ is $\mathbb{Q}$-Cartier, $K_V + B_Z|_V + D$ is $\mathbb{R}$-Cartier, and we define the $\mathbb{R}$-divisor D' on V' by

$$(\mu \times \mathrm{id}_{\mathbb{A}^1})^*(K_V + B_Z|_V + D) = K_{V'} + \mathrm{pr}^* B_Y|_{V'} + D'.$$

By (8.7),

$$D' = \mathrm{pr}^*\, q^* M_W|_{V'} + \sum_{i\in I} \nu_i \,\mathrm{div}_{V'}(\rho_i) \sim_{\mathbb{R}} \mathrm{pr}^*\, q^* M_W|_{V'}.$$

Since $\mu \times \mathrm{id}_{\mathbb{A}^1}$ is an isomorphism outside $E \times \mathbb{A}^1$, there exists a real number α with

$$D' = \widetilde{D} + \alpha(E \times \mathbb{A}^1)|_{V'}, \quad \text{where } \widetilde{D} := (\mu \times \mathrm{id}_{\mathbb{A}^1})_*^{-1} D \geq 0.$$

Let $B_{V'} := \mathrm{pr}^*\, B_Y|_{V'} + D'$. Then $a(v, V, B_Z|_V + D) = a(v, V', B_{V'})$ for any prime divisor v over V', and if $\mathrm{center}_{V'}\, v \subset E \times \{0\}$, then $\mathrm{center}_V\, v = \{z\}$.

Let $C := \{\tau \in \mathbb{A}^1 \mid (o, \tau) \in V\}$, an open neighborhood of 0, and for $\tau \in C$, $E \times \{\tau\} \subset V'$ is the fiber of $\mu \times \mathrm{id}_{\mathbb{A}^1}$ over (o, τ) and is identified with W by q. For general $\tau \in C$, $E \times \{\tau\} \not\subset \mathrm{Supp}\, \widetilde{D}$, and restricting the $\mathbb{R}$-linear equivalence class of D' to $E \times \{\tau\}$, since $(E \times \mathbb{A}^1)|_{E\times\{\tau\}} \sim -A$, we obtain

$$\widetilde{D}|_{E\times\{\tau\}} \sim_{\mathbb{R}} M_W + \alpha A. \tag{8.8}$$

If $\alpha < 0$, then $M_W \sim_{\mathbb{R}} \widetilde{D}|_{E\times\{\tau\}} + (-\alpha)A$ with $\widetilde{D}|_{E\times\{\tau\}} \geq 0$ and $(-\alpha)A$ ample, so M_W is big, contradicting **Step 1**. Thus $\alpha \geq 0$.

Suppose that $\alpha > 0$. Let w_0 be a local equation of $E \times \mathbb{A}^1$ at the generic point of $E \times \{0\}$. Then w_0, s form a regular system of parameters there. For any prime divisor P_W on W, $q^* P_W \not\supset E$, so the only component of $\mathrm{pr}^*\, B_Y$ containing $E \times \{0\}$ is $E \times \mathbb{A}^1$, of coefficient 1. For any positive integer k, let v_k be the monomial valuation in w_0, s with weights $(k, 1)$. Then v_k is divisorial and $\mathrm{center}_{V'}\, v_k = E \times \{0\}$. Since $\mathrm{mult}_{E\times\mathbb{A}^1} B_{V'} = 1 + \alpha$ and $\widetilde{D} \geq 0$,

$$a(v_k, V', B_{V'}) \leq k + 1 - (1 + \alpha)k = 1 - k\alpha.$$

Suppose that $\alpha = 0$. By (8.8) and **Step 1**, $p \in \mathrm{Supp}(\widetilde{D}|_{E\times\{\tau\}})$ for general $\tau \in C$, where we regard p as a point of $E \times \{\tau\}$. Since $\mathrm{Supp}\, \widetilde{D}$ is closed in V', it contains the curve $\{p\} \times C$. Let $\beta > 0$ be the sum of the coefficients in $\widetilde{D}$ of the components of $\widetilde{D}$ containing $\{p\} \times C$. At $(p, 0)$, let w_0 be a local equation of $E \times \mathbb{A}^1$ and w_j a local equation of $\mathrm{pr}^*\, q^* R_j$ for $j = 1, \ldots, 5$. Since $R_1, \ldots, R_5$ intersect transversely at p, $w_0, \ldots, w_5, s$ form a regular system of parameters of V' at $(p, 0)$, and $\{p\} \times C = (w_0 = \cdots = w_5 = 0)$ near $(p, 0)$. Since $p \notin \mathrm{Supp}\, F$, near $(p, 0)$ we have

$$B_{V'} = \sum_{j=0}^{5} (w_j = 0) + \widetilde{D}.$$

For any positive integer k, let v_k be the monomial valuation in $w_0, \ldots, w_5, s$ with weights $(k, \ldots, k, 1)$. Then v_k is divisorial and $\mathrm{center}_{V'}\, v_k = \{(p, 0)\} \subset E \times \{0\}$. A local equation of any component of $\widetilde{D}$ containing $\{p\} \times C$ is in the ideal $(w_0, \ldots, w_5)$, so has value $\geq k$ under v_k. Thus

$$a(v_k, V', B_{V'}) \leq 6k + 1 - 6k - k\beta = 1 - k\beta.$$

For $\alpha > 0$ and for $\alpha = 0$, letting $k \to +\infty$, we obtain $\mathrm{mld}(V \ni z, B_Z|_V + D) = -\infty$. Thus (4) holds. If Conjecture 1.2 held for $d = 2$ and $\epsilon = 1$, then by (1)–(3) there would be a real number $\delta > 0$ and an $\mathbb{R}$-divisor $D \geq 0$ on Z with $D \sim_{\mathbb{R}} M_Z$ and $\mathrm{mld}(Z \ni z, B_Z + D) \geq \delta > 0$, contradicting (4) with $V = Z$. □

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School of Mathematical Sciences, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China

Beijing International Center for Mathematical Research, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China
Email address: liujihao@math.pku.edu.cn